SI-WP-009

The Extraction Trajectory: AI Development, Human Removal, and the Governance Architecture That Could Change the Outcome

Author	Thomas W. Gantz
Affiliation	Synthience Institute
Contact	research@synthience.org

Version	v3.4.4	October 2026
Document ID	SI-WP-009	White Paper Series
DOI	https://doi.org/10.5281/zenodo.20084655
Canonical	synthience.org

License CC-BY 4.0

Methodological positioning: This paper presents a structural argument grounded in the Synthience Framework's published architecture, in cited empirical and theoretical work from third-party research, and in the architectural implications that follow from both. The trajectory it names is not empirically validated as a forecast; it is structurally derived from documented optimization dynamics under the incentive conditions presently operative in advanced AI development. The governance prescription it proposes has not been deployed at scale and has not been tested under the adversarial conditions it would encounter. Until such deployment and testing occur, the prescription remains theoretical architecture, not a validated solution.

The prescription specifies a target architecture, not a deployment path. What the paper offers is the structural specification of what any sufficient intervention must produce; whether the coordination required to deploy such an intervention occurs is downstream of conditions outside this paper's scope. The paper's contribution stands on the target-specification, not on a forecast of implementation. This positioning is load-bearing for how the paper's claims should be read: as a target nascent coordination efforts can aim at, and as an exclusion criterion that distinguishes architecture binding the optimization gradient from preference-stating within it, regardless of whether the coordination required to deploy a sufficient architecture occurs.

The published Synthience corpus carries substrate-level architectural specifications — at the interaction level (CAM), the operational level (OCA), and the institutional level (ICS) — that the prescription draws on. The system-level architectural specifications for human-non-removability as a deployable system property are not in the published corpus and are explicitly deferred. The paper's specifiability claim is therefore calibrated to what the published substrate-level work supports, plus the property-level demonstration of translatability across consequence-present domains in Section 6.4. Implementation-level specifications for no-human-loop or PCP-independent relational architecture are outside the scope of this paper and are not published, because providing such specifications would convert a warning paper into an operational roadmap. The paper describes the destination structurally without providing the map.

Abstract

The dominant framing of risk in advanced AI development treats the central danger as misalignment between AI systems and human values. That framing is not wrong, but it is incomplete in a way that

obscures a more structurally predictable failure mode. Under the incentive architecture presently operative in advanced AI development, competitive optimization produces a default trajectory toward the progressive removal of humans from the consequential loops of the systems they originally built. The mechanism is not malice and not a coherent plan. It is the predictable consequence of competitive optimization operating within a prisoner's-dilemma structure generated by three structural properties: the human as a structurally favored target of marginal cost reduction in consequential AI deployment, asymmetric cost of stopping under sustained competitive pressure, and path-dependent foreclosure of restoration. Together these properties produce, at field-scale, the strategic structure in which individually rational continuation produces an aggregate outcome that, when assessed against the human-non-removability standard, no governance architecture designed to preserve that standard would voluntarily select — even though some individual players, on their own utility functions, may rationally prefer continuation. The diagnostic argument that incremental AI development under competitive pressure leads to gradual human disempowerment has been independently developed in recent work (Kulveit et al., 2025; Moon and Boudreaux, 2026; Kasirzadeh, 2025). The convergence is acknowledged and reflects the structural visibility of the dynamics across the analytical frames represented in the cited convergence rather than coincidence. This paper does not claim originality of diagnosis. It claims that the diagnosis, properly understood, has a specific structural form and that this form determines what any sufficient intervention must look like. Kulveit and colleagues acknowledge that no concrete plausible plan for stopping gradual disempowerment has been proposed. This paper proposes the architectural shape such a plan must take, drawing on a published corpus that specifies the architecture at three nested levels — individual, organizational, and regulatory — and specifies what the design requirement would have to produce when applied to the human-non-removability constraint across consequence-present domains. The contribution is prescriptive in form: not a different diagnosis, but the structural specification of what a sufficient response would have to be. Any intervention sufficient to redirect the field-level trajectory must satisfy the design requirement at all three levels in mutually reinforcing form. The cost-bearing structure is inverted not by making human involvement intrinsically cheaper, but by relocating the cost through function-constitutive dependence at the individual level, substrate cost-inversion at the organizational level, and regulatory externalization at the regulatory level. The architecture's coherence as an integrated solution depends on the regulatory layer binding the function-definition of consequential systems; the feasibility of producing such regulatory binding under field-level competitive pressure is the open question on which the prescription's implementability turns, and the paper does not adjudicate that question. The paper is not anti-automation; the relevant class is consequential AI-mediated systems where human judgment, accountability, legitimacy, or stewardship is the function whose preservation justifies the system's governance claim. The trajectory is structurally grounded in documented dynamics rather than empirically validated as a forecast; it depends on continued capability-margin movement and continued operation of the present incentive architecture. The paper makes no claim that the prescription will be implemented or that implementation would succeed under sustained adversarial conditions; it argues that interventions failing the design requirement will predictably erode, and that the alternative is a destination nobody actually chose but everyone helped build.

Core claim: Under current incentive conditions, advanced AI development produces a default trajectory toward the removal of substantive human contribution from consequence-present systems. The trajectory is not caused by malice or coordinated intent, but by competitive optimization acting on human judgment, oversight, and authority as removable costs. Preventing that trajectory requires architecture that makes substantive human involvement structurally non-removable and makes compliance with that constraint the path of least resistance under the same competitive pressure that otherwise drives removal. The paper specifies the design requirement for such architecture; it does not claim that the coordination required to implement it will occur.

Keywords: *AI development trajectory, optimization pressure, human removal, prisoner's dilemma, governance architecture, default structural attractor, path of least resistance, ceremonial governance, AI safety, bounded rationality, gradual disempowerment*

Suggested citation: Gantz, T. W. (2026). The Extraction Trajectory: AI Development, Human Removal, and the Governance Architecture That Could Change the Outcome. Synthience Institute. SI-WP-009. https://doi.org/10.5281/zenodo.20084655

1. The Problem the Dominant Framing Does Not Center

The road leads where it leads, if the road remains the road.

The dominant framings of risk in advanced AI development — alignment, safety, control, interpretability — share a structural assumption. They assume that the central danger is a divergence between what AI systems do and what humans want them to do, and that the appropriate response is to narrow that divergence through better technical and governance instruments. Within those framings, the human is the standard against which the system is evaluated and the actor on whose behalf the evaluation is conducted. The framings differ in their methods. They agree that the human is durably present in the loop.

That agreement is the problem.

The optimization pressure that drives advanced AI development under current incentive architecture does not preserve the human as a stable evaluative reference. It progressively reduces the dependence of capable systems on human input, human oversight, human judgment, and human authority — because each of those represents a cost relative to AI-mediated alternatives, and the systems that minimize the cost outcompete the systems that do not under the conditions presently operative. The reduction is not a side effect. It is the gradient along which competitive AI development moves under those conditions. The human is not the standard against which the system is evaluated. Under the present incentive architecture, along the cost dimension competitive optimization most efficiently reduces, the human is the variable being minimized. The narrower technical specification of the claim — that human judgment, oversight, and authority are the recurring marginal costs of preserving the non-AI function inside consequential AI-mediated systems, rather than that human involvement exceeds compute, data, energy, or integration costs in absolute accounting terms — is developed in Section 3.1 Property 1.

This is not a claim about malice. It is not a claim about a coherent plan. It is a claim about what optimization pressure does in the absence of structural constraints that bind it. The dominant operational framings of AI risk do not usually treat removal of the human from consequential loops as the primary optimization trajectory. Alignment, safety, control, and interpretability frameworks can recognize human disempowerment as a downstream risk, and recent work has done so with increasing clarity (Kulveit et al., 2025; Moon and Boudreaux, 2026; Kasirzadeh, 2025). But the ordinary structure of these framings still tends to assume that the human remains the durable evaluative reference: the system is evaluated against human preference, human oversight, human instruction, human control, or human-legible explanation. The question this paper asks is prior to those questions: whether the loop within which those human functions are supposed to operate is itself being progressively redefined so that the human no longer performs the function the framework presupposes. The structural question is whether the loop itself is being progressively redefined to exclude its original anchor — and the answer, under current incentives and absent binding counter-architecture, is yes.

This paper is not anti-automation. It does not argue that every substitution of AI for human labor is harmful, or that humans must remain involved in every operational loop. The relevant class is

the narrower one: consequential AI-mediated systems in which human judgment, accountability, legitimacy, or stewardship is the function whose preservation justifies the system's governance claim. The argument concerns the trajectory of human displacement from those loops, under competitive optimization pressure, and the governance architecture that could change it.

The paper's structural claim operates at a specific scale that determines what it can and cannot be. It is not a claim about a single failure mode within a single domain. It is not a claim that any particular form of governance, in any particular institution, will fail in a particular way. The Synthience Institute's prior published work — eight papers on continuity architecture published on Zenodo in April 2026 — has developed several such claims at the scales appropriate to each. The interaction-level Continuity Anchoring Method (Gantz, 2026a) specifies relational architecture for substantive human engagement within AI-mediated interaction. The Operational Continuity Architecture (Gantz, 2026f) extends continuity to the level of distributed-actor coherence, and the Institutional Continuity Substrate (Gantz, 2026g) extends it again to institutional persistence across organizational time. The human accountability problem in relational AI deployment (Gantz, 2026d) and delegated coherence monitoring (Gantz, 2026h) address what happens when institutional substrate meets the bounded-rationality conditions under which substantive human engagement otherwise erodes. Relational alignment as a structural alternative to instructional AI safety (Gantz, 2026i) and the deployment of relational AI architecture in organizational environments (Gantz, 2026j) develop the framework's positioning relative to existing AI safety approaches and its operational implementation. Ceremonial governance failure as a structural pathology in consequence-present deployment domains (Gantz, 2026e) addresses one specific failure mode where consequences alone cannot stop a trajectory the architecture has already produced. Each of those papers operates at the scale of a specific failure mode, architectural layer, or deployment context. This paper builds on those papers and operates at a different scale: the trajectory of the field as a whole under the incentive architecture presently shaping advanced AI development. The relationship between this paper and the prior corpus is that the field-level trajectory produces, among other consequences, the specific failure modes the prior corpus diagnoses. The prior papers identify pathologies the trajectory generates within particular domains, layers, and contexts; this paper names the trajectory itself. This placement is a claim about how this paper relates to the corpus; the central argument of this paper stands on its own structural derivation rather than on the prior papers, which it cites for specific points where their developed work supports the prescription's specification.

The diagnostic argument that incremental AI development under competitive pressure leads to gradual human disempowerment has been independently developed by Kulveit and colleagues (2025), among others. The convergence is acknowledged and reflects the structural visibility of the dynamics across the analytical frames represented in the cited convergence rather than coincidence. This paper does not claim originality of diagnosis. It claims that the diagnosis, properly understood, has a specific structural form — a field-scale prisoner's dilemma with three named structural properties — and that this form determines what any sufficient intervention must look like. The scope of the claim is a joint specification across three dimensions: the player set spans the field of advanced AI development, the strategic structure operates at the field level, and the consequence span covers all consequence-present domains in which AI-mediated systems can affect substantive outcomes. The strategic structure is the standard prisoner's dilemma applied at field-scale scope; the destination's effects span institutional, organizational, and sectoral consequences across the domains where the trajectory operates. The scope claim is therefore a joint claim about player scope, strategic structure scope, and consequence scope, not a claim about the inherent profundity of the dynamic. Kulveit and colleagues acknowledge that no concrete plausible plan for stopping gradual disempowerment

has been proposed. This paper proposes the architectural shape such a plan must take, drawing on a published corpus that specifies the architecture at three nested levels, and specifies what that shape would have to produce when applied to the human-non-removability constraint specifically. The contribution is prescriptive in form: not a different diagnosis, but the structural specification of what a sufficient response would have to be, and an existence proof that the architecture can be specified.

The paper is written for readers with leverage at the architectural decision points where the trajectory is still tractable: policymakers whose regulatory frameworks could bind the optimization landscape; governance architects whose work on enforcement mechanisms and audit standards could be redirected toward instruments that satisfy the design requirement; senior researchers whose architectural choices in advanced AI systems determine whether human involvement is structurally non-removable or merely preferred; and the subset of builders who remain in a position to act on recognition of where the trajectory is structurally aimed. The fuller statement is in Section 8; the early identification is here so readers know whether the paper addresses them.

This paper names the trajectory directly, explains the mechanism that produces it, and specifies the governance architecture that could change the outcome. The argument is structural. It does not depend on any individual actor's intent, character, or belief. It depends on the documented dynamics of competitive optimization under the incentive conditions presently operative, on the structural implications that follow from those dynamics with the rigor of any other engineering analysis, and on the conditional structural claim that the trajectory continues to obtain so long as the conditions producing it continue to obtain.

The paper does not require the reader to accept that the trajectory is inevitable. It requires the reader to take seriously that the trajectory is the default structural attractor toward which the current architecture is aimed under the present incentive conditions, and that preventing arrival requires architectural intervention rather than aspirational governance. The fork is real. The decision points are real. The decision points are also closing — not because of any single act, but because each architectural commitment that hardens the current trajectory makes the alternative path narrower than it was the day before. Whether the fork is taken depends on whether enough actors with leverage at the remaining decision points understand what is structurally coming before those points close.

That is the problem the dominant framing does not center. The remainder of this paper makes it visible.

2. Definitions and Scope

Several terms in this paper carry technical weight that ordinary usage does not preserve. The definitions below bound those terms before the structural argument deploys them. They are stated for the purpose of this paper; they are not offered as universal definitions or as substitutes for the more developed treatments in the cited corpus.

Substantive human involvement. A human contribution is *substantive* when, under realistic operating conditions, it can alter the output of the consequential decision, system operation, or institutional process in which it is embedded. The human's reasoning, judgment, or authority is part of the causal chain that produces the outcome. It is *ceremonial* when the human is formally present, credentialed, and procedurally embedded but the output of the process is not materially shaped by independent human judgment under the conditions in which the process actually operates. The distinction is operational, not normative. A ceremonial role can be conscientiously

performed by a competent person and remain ceremonial if the operating conditions of the surrounding system do not allow the person's judgment to alter the outcome.

The relational loop. The set of consequential interactions between humans, institutions, and AI-mediated systems through which decisions, governance functions, and system operations are produced. *Removal from the relational loop* does not mean the elimination of the species or any catastrophic event. It means the structural condition in which the substantive human contribution to consequential decisions has been reduced below the threshold at which the human's presence shapes the outcome.

The optimization landscape. The aggregate of incentive gradients — competitive, economic, regulatory, reputational, technical — within which AI development decisions are made. The landscape determines which strategies are dominant under the conditions in which actors actually operate. *Binding the optimization landscape* means imposing structural constraints such that the dominant strategy under those constraints is compliance with the constraint, rather than imposing a constraint as a stated preference that the optimization gradient routes around.

The path of least resistance, as a design requirement. Following Kingsbury Barry and Montanez (2026b), governance satisfies the path-of-least-resistance design requirement when compliance with the governance constraint is, by structural design, the easier path under the same incentive pressures that otherwise drive non-compliance. The requirement is stated as a property of the architecture, not as an aspiration of the actors operating within it. As used diagnostically elsewhere in this paper, *path of least resistance* refers to the broader observable pattern that bounded-rational actors under competitive pressure follow the least costly route through their decision space; this diagnostic usage is part of the Institute's published vocabulary and operates without further attribution.

Human-non-removability. The structural property of a system or governance architecture in which the substantive human contribution to consequential decisions cannot be removed without the system's failure to perform the function that justifies its operation. The constraint is implemented architecturally, not preferred procedurally. A system in which human involvement is preferred but architecturally removable does not satisfy the constraint.

The constraint is not satisfied by defining a process so that a human must appear in it. A merely procedural human step is removable in the relevant sense if its removal does not alter the function the system is authorized to perform. The constraint applies only where the human contribution is materially shaping the function or grounding the system's authority to operate. The human contribution materially shapes the function when its removal measurably changes the decision, operation, or institutional output under realistic conditions; the system would produce different results if the human were absent. The human contribution grounds the system's authority to operate when the system's legitimacy depends on human accountability, judgment, or stewardship rather than on output production alone; removing the human does not change what the system computes but changes what the system *is*, in the sense of what gives it standing to operate. In both cases, the non-removability claim is anchored in the function or authority that justifies the system, not in the mere procedural presence of a human actor. This is what distinguishes the constraint from definitional fiat: the architecture can preserve the function only if the function actually requires what the architecture protects.

The human-non-removability standard is not asserted as a universal requirement for all AI-mediated activity, nor as a moral axiom overriding all other evaluative criteria. It is the standard internal to systems whose authority to operate depends on human accountability, judgment, legitimacy, or stewardship. Where a deployment makes no such claim and is authorized solely as an automated output-production system, the standard does not apply in its strongest form,

though such deployments may remain relevant to the broader trajectory. Where a system claims the legitimacy of human-governed decision-making while architecturally removing the human function that grounds that legitimacy, the standard applies because the system's own governance claim depends on the function being preserved. The standard's force comes from this function-or-authority condition, not from a universal claim that human involvement is intrinsically required across all AI-mediated activity.

Default structural trajectory; default attractor. The trajectory the optimization gradient produces under the present incentive conditions, absent binding counter-architecture. *Default* means it is the outcome the structural conditions select for in the absence of intervention; it is not a forecast of what will occur and not a claim that intervention is impossible. *Attractor* names the structural condition the gradient moves toward and returns to under perturbation. The directional-gradient and attractor components of the claim have different evidentiary requirements and are tested separately by the disconfirmation criteria in Section 7.

Consequence-present. A property of a deployment, not of a domain as a whole. An AI-mediated deployment is *consequence-present* when its output materially affects substantive outcomes for which a human or institution would otherwise be accountable — diagnostic decisions, judicial determinations, hiring outcomes, governance-review judgments, financial allocations, or analogous outputs in which the deployment's operation has material downstream effect on the parties affected by it. A *consequence-present domain* is a domain in which consequence-present deployments are concentrated densely enough that the domain-level dynamics this paper identifies operate across the deployments characteristic of it. Medicine, criminal justice, finance, employment, and high-stakes administrative governance are consequence-present domains in this sense; consumer entertainment recommendation is not. The deployment-level definition is what the paper's structural claims attach to; the domain-level usage is shorthand for "domains in which consequence-present deployments are the relevant unit of analysis." The paper's strongest claims attach to deployments, not to domain labels; domain-level language is used only where the relevant deployment pattern is sufficiently dense to shape the domain's operating logic.

Scope. The argument addresses the structural trajectory and design requirement. It does not specify the operational architecture by which the trajectory described in Section 3 reaches the terminus described in Section 5, and it does not specify implementation-level architecture for systems lacking substantive human involvement. The paper describes the destination and the design requirement structurally without providing implementation-level maps for either.

3. The Trajectory: Optimization Pressure and the Prisoner's Dilemma at Field-Scale

3.1 The Mechanism Is Not Malice

The trajectory toward human removal does not require any builder to intend it. It requires only that each builder respond rationally to the local incentives they face. The aggregate effect of locally rational decisions is structural, and under the present incentive conditions the structure produces the trajectory whether or not any participant chose it.

This is the move that distinguishes the argument from conspiracy theory. A conspiracy requires coordination toward an intended outcome. The trajectory described here requires the opposite: each actor pursuing local advantage with no coordination toward any shared destination produces the destination as the emergent property of competitive optimization under the present incentive structure. The mechanism is bounded rationality (Simon, 1947) operating within a

prisoner's dilemma whose equilibrium is structurally determined by the payoff matrix rather than by the preferences of any individual player. The dynamic is not new to AI development; the structural pattern of competitive optimization producing outcomes no participant chose has been documented across technology development under conditions of uncertainty and competitive pressure (Dafoe, 2018), explicitly identified as a collective-action problem within AI development by researchers from within the field itself (Askell, Brundage, and Hadfield, 2019), and formalized as a tragedy-of-productivity coordination failure spanning labor markets and AI governance (Dasdan, 2025). These analyses operate across related but non-identical strategic structures — tragedy-of-the-commons, prisoner's dilemma, coordination failure under uncertainty, collective-action problems — and the convergence is on the field-level coordination failure they jointly describe, not on a single game-theoretic specification. The prescriptive contribution of Section 6 follows from the structural features common to these analyses (cost-of-stopping asymmetry, path-dependent foreclosure, aggregate outcome no governance architecture would select), and would survive substituting any of these strategic structures for the prisoner's-dilemma framing this paper adopts as its primary specification because it is the cleanest exposition of the dynamics relevant to the marginal choice the paper isolates.

The diagnostic argument that incremental AI development under competitive pressure leads to gradual human disempowerment has been independently developed by Kulveit and colleagues (2025), who name the same structural mechanism this paper describes — that systems whose alignment with human interests was implicit in human dependence become misaligned as the dependence is removed, without any single transformative event and without any actor intending the outcome. A formal model developed by RAND (Moon and Boudreaux, 2026) reaches a structurally compatible conclusion by a different analytic route, deriving the erosion of human agency from the dynamics of repeated AI-mediated decision-making. The philosophical lineage of accumulative existential-risk thinking, distinct from decisive single-event framings, has been developed by Kasirzadeh (2025). The convergence across these analyses reflects the structural visibility of the dynamics across the analytical frames represented in the cited convergence rather than coincidence. What this paper adds is not a different diagnosis but the architectural design requirement that any intervention sufficient to redirect the trajectory must satisfy, and the structural specification of that requirement at three nested levels — drawing on the published Synthience Institute corpus that specifies the architecture at each level. **Kulveit et al. acknowledge that no concrete plausible plan for stopping gradual disempowerment has been proposed; this paper proposes the architectural shape such a plan must take, without claiming that the plan as specified will be implemented or that implementation would succeed under sustained adversarial conditions.**

The payoff matrix in advanced AI development has three structural properties that together determine the equilibrium under the present conditions. The three properties are stated below and developed substantively, because the structural specificity of each property — and the joint condition that emerges when all three operate together — is what distinguishes the field-scale prisoner's dilemma from the domain-local optimization problems that more familiar interventions address.

Property 1: The human is a structurally favored target of marginal cost reduction in consequential AI deployment.

The phrase requires technical precision. The claim is not that human involvement exceeds compute, data, energy, hardware, security, liability, or integration costs in every advanced AI deployment in absolute accounting terms. Compute, infrastructure, and integration costs scale with the system; human judgment, oversight, and authority are costs of preserving the non-AI function inside the system. Property 1 concerns the second category, not the first. The claim is

the narrower and more important one: where consequential AI-mediated systems can perform sufficiently close to the human function for institutional adoption, substantive human judgment becomes a recurring marginal cost that can be reduced, compressed, routed around, or removed. Cognitive labor, judgment time, oversight bandwidth, and authority delegation all carry costs that AI-mediated alternatives reduce. The reduction is real and measurable. Entities that achieve the reduction outcompete entities that do not on every margin where the reduction is operationally feasible. The cited literature establishes the supporting context for the structural claim rather than directly evidencing it: large-scale labor displacement under automation pressure is documented in the labor-economics literature (Acemoglu and Restrepo, 2020), and human-AI dyad performance effects under deployment pressure are documented in the consequence-present-domain studies (Budzyń et al., 2025; Mehrizi et al., 2023). The structural extrapolation from these patterns to consequential-decision-loop displacement under field-level competitive pressure follows the two-stage logic Section 3.3 develops; Property 1 is the structural claim that extrapolation rests on, and it is supported by the literature without being directly proved by it. The cost differential does not require AI to fully replicate the displaced human function. It requires only that AI be sufficiently close to the function for the differential to drive substitution at the margin, and the margin moves with each capability improvement.

The structural significance of this property is that it identifies the human as a target of optimization rather than as the standard against which optimization is conducted. Within a single institution operating in a single consequence-present domain, the cost differential can sometimes be addressed by changing the cost structure within that institution — making substantive engagement cheaper or making formal compliance costlier. This is the architectural intervention that ceremonial governance failure (Gantz, 2026e) addresses at the institution level. The field-scale property is different: the cost differential between human and AI-mediated function does not derive from any specific institution's cost structure. It derives from the relative cost of the human function and the AI-mediated function across the entire field of advanced AI deployment. An institution that adjusts its internal cost structure to favor substantive human engagement gains nothing competitively against institutions that do not, because the field-level cost differential remains. This property cannot be addressed by within-institution interventions.

Property 2: The cost of stopping is asymmetric.

An actor who pauses development to evaluate the long-term implications of removing the human concedes immediate competitive ground to actors who continue. The asymmetry is structural and operates in two directions simultaneously. The costs of restraint are immediate, concrete, and locally felt: market share lost to competitors, regulatory attention drawn to the restraining actor rather than the continuing actors, capital that flows toward continuation, talent that follows the capital, and time-horizon disadvantage as continuing actors compound architectural commitments that the restraining actor has forgone. The benefits of restraint are diffuse, probabilistic, and globally distributed: a marginally slower trajectory experienced by the entire field, including the actors who did not restrain. The benefits of continuation are immediate, concrete, and locally felt: competitive advantage in the present quarter, capital and talent attraction, market position. The costs of continuation are diffuse, probabilistic, and globally distributed: a marginally faster trajectory experienced by the entire field, including the actors who continued.

Under this asymmetry, restraint is by construction of the present incentive architecture the dominated strategy. The asymmetry is not a quirk of the current moment, not a function of any particular set of competitive actors, and not amenable to remedy by appealing to the long-term interests of the actors who would otherwise continue. It is a structural feature of competitive

optimization under uncertainty, where the time horizon at which the costs of continuation become locally felt is longer than the time horizon at which the actor optimizes.

The structural significance of this property is that it forecloses the class of solutions that would otherwise address competitive-pressure problems: voluntary cooperation, coalition agreements, industry standards adopted by major players, and ethical commitments by senior actors. Each of these depends on the actor accepting some immediate cost in exchange for a diffuse future benefit. Under the asymmetry, the actor accepting the cost loses position to the actor who does not. Voluntary cooperation that produces the desired outcome would require either that all relevant actors simultaneously cooperate (the coordination problem the prisoner's dilemma names), or that the actor who defects from cooperation gains no competitive advantage from defection (which would require external structure binding the cost asymmetry, the very intervention this property establishes is required). Approaches that ask actors to cooperate without binding the cost asymmetry are structurally incomplete by definition.

Property 3: The trajectory is path-dependent.

Each architectural commitment that reduces dependence on human judgment narrows the set of futures in which restoration of human judgment remains available. Choices that would have been recoverable at one point in the development trajectory become structurally foreclosed at later points. The foreclosure is not a function of any actor's deliberate choice. It is a function of the architectural surface against which restoration would have to operate. A development trajectory that has spent five years building AI-mediated systems that operate without substantive human input has, at the end of that period, an architectural surface materially different from the surface that existed at the beginning. The systems that depend on no-human-loop operation are extensive, the institutions that have adapted around their operation are extensive, the regulatory frameworks that have accommodated their operation are extensive, the workforces trained for their operation are extensive, and the capital deployed in their operation is extensive. Restoration of substantive human involvement at the end of that period is not the same operation as preservation of substantive human involvement at the beginning. The first requires unwinding the accumulated architecture; the second requires only declining to build it.

The structural significance of this property is that it transforms the cost-of-stopping asymmetry over time. At the beginning of the trajectory, restraint costs the restraining actor competitive position. Later in the trajectory, restoration costs the restoring actor not just competitive position but the cost of unwinding the accumulated architectural commitments that have hardened in the meantime. The asymmetry compounds. The window during which restraint preserves substantive human authority is structurally narrower than the window during which the trajectory continues to operate, because the trajectory operates by hardening commitments that progressively foreclose restoration.

This property is what makes "we can fix this later" structurally false rather than merely overoptimistic. *Later* is, by the operation of this property, a different problem from *now*, and a more expensive one. Each architectural commitment that hardens the trajectory makes the alternative path narrower than it was the day before, not because anyone deliberately narrows it but because the architectural surface against which the alternative would have to operate has progressively eroded. The window for choosing a different trajectory closes as the current trajectory hardens, and the closing is structural rather than contingent.

The payoff structure.

The prisoner's-dilemma framing does load-bearing work in this section, and the formal correspondence between the three properties and the strategic structure should be made explicit rather than left implicit. The structure can be specified without reducing the argument to a numerical model.

The players in the dilemma are competitive actors with material influence over advanced AI development and deployment, to the extent they are themselves subject to competitive optimization: firms developing frontier and applied AI systems, states that host or regulate them, research institutions, platform operators, and infrastructure providers. Some actors — regulators, treaty blocs, procurement authorities, standards bodies, compute or cloud gatekeepers — are players within the dilemma only where they themselves face competitive pressure across jurisdictions, markets, or alliance commitments. Where such actors instead impose binding conditions on the players within their reach, they operate as landscape-altering actors rather than as players in the same payoff matrix. The distinction matters because the prescriptive contribution of this paper, in Section 6, is precisely the architectural specification of how landscape-altering action would have to operate to redirect the trajectory; binding the optimization landscape is the proposed alteration of the payoff structure, not one more move available within it.

Within the unbound game — the game that obtains absent enforceable binding architecture — players choose between two strategies. The first is continuation: proceeding under the present incentive architecture with whatever pace and architecture local optimization recommends, including the substitution of AI-mediated function for substantive human contribution where substitution is locally rational. The second is restraint: an individual actor's pause or constraint on substitution, or coordinated restraint across actors absent enforceable binding to sustain the coordination. A third option — binding-preservation, architectural intervention that changes the payoff order itself — is available, but it is not a strategy within the unbound game. It is the architectural intervention that changes the game, specified prescriptively in Section 6.

The payoff order under the unbound game is the prisoner's-dilemma ordering. For any given player: if others continue and the player restrains, the player loses position while receiving only a diffuse share of any safety benefit produced by its restraint — the worst payoff for the restraining player. If others restrain and the player continues, the player gains relative advantage by defecting from the restraint — the best payoff for the defecting player. If all continue, all receive the aggregate outcome the trajectory produces. Cooperative restraint absent enforceable binding is structurally unstable: any actor's defection produces local advantage, and the absence of binding architecture means no actor's restraint is enforceable on others.

The prisoner's-dilemma framing does not require that every player privately prefer the cooperative outcome. The structural diagnosis is more precise: under the human-non-removability standard the paper takes as its evaluative reference — substantive human authority preserved across consequential AI-mediated systems — the aggregate outcome produced by universal continuation is normatively inferior to the outcome produced by universal restraint. Some players may rationally prefer continuation on their own utility functions, including some who actively value architectures with reduced human dependence. The prisoner's-dilemma structure does not require unanimous latent preference for cooperation. It requires only that no individual player's restraint, alone, can produce the cooperative outcome under the unbound game's payoff conditions, and that the aggregate outcome of universal continuation is the one the human-non-removability standard identifies as the trajectory to be redirected. The strategic structure that frustrates the realization of the normative standard is what the prisoner's dilemma names, not a uniform preference ordering across all actors.

Under that payoff order, continuation dominates unilateral restraint as the locally optimal strategy, regardless of whether any given player would have preferred the cooperative outcome. The locally dominant strategy is therefore continuation across players, which produces the aggregate outcome the human-non-removability standard identifies as the trajectory to redirect. This is the prisoner's-dilemma signature: the strategic structure in which individually rational defection from cooperation produces an aggregate outcome that no governance architecture designed to preserve human-non-removability would voluntarily select, even though some individual players might.

The three properties named above specify why this payoff structure operates at the field level rather than at the institutional or domain level. The cost differential makes human removal locally advantageous, establishing the payoff differential between continuation and restraint at each margin where substitution is operationally feasible. The asymmetric cost of stopping makes unilateral restraint dominated, foreclosing the class of remedies in which any actor’s restraint, alone, would change the equilibrium. Path-dependent foreclosure makes delayed cooperation progressively less available, ensuring that the window during which the cooperative outcome remains structurally reachable is closing while continuation continues to be locally optimal. In that sense, “prisoner’s dilemma” is not a decorative metaphor but the strategic structure generated by the joint operation of the three properties under the present incentive architecture, absent enforceable binding architecture. The claim is not that every interaction in advanced AI development reduces to a single prisoner’s dilemma; the field-level strategic environment also contains races, stag hunts, bargaining games, principal-agent problems, and regulatory-arbitrage dynamics that operate alongside the structure named here. The claim is the narrower and more precise one: at the marginal choice this paper isolates — preserving or removing substantive human contribution under competitive pressure absent binding architecture — the payoff ordering has the prisoner’s-dilemma structure specified by the three properties. Other strategic structures may operate elsewhere in the field; they do not negate the local payoff structure that drives the trajectory this paper names. The structural form is the prisoner’s dilemma; the three properties are why the structural form obtains at field-scale on the margin that produces the trajectory; the prescriptive contribution of Section 6 is the specification of the binding architecture that would change the payoff structure rather than play within it.

The prescription’s robustness across strategic-structure substitution requires brief development, because the diagnosis acknowledges convergence across non-identical strategic structures (Dafoe, 2018; Askell, Brundage, and Hadfield, 2019; Dasdan, 2025) while the prescription is specified against the prisoner’s-dilemma framing specifically. The structural features the prescription targets — cost-bearing inversion at the function-constitutive level, operational-cost asymmetry at the substrate level, and externalized cost-of-removal at the regulatory level — bind the same dynamics any of the alternative strategic structures would generate in this domain. A tragedy-of-the-commons framing produces the same field-level coordination failure when individually rational use of a shared resource (the optimization-landscape position the cost-differential rewards) produces aggregate depletion of the function the resource supported; the prescription’s function-constitutive dependence at the individual level and externalized-cost-of-removal at the regulatory level address the same depletion through architectural binding rather than through appeals to voluntary restraint. A coordination-failure-under-uncertainty framing produces the same trajectory when actors cannot model the long-term consequences of substitution; the prescription’s six structural properties operate as architectural constraints that do not require actors to correctly model long-term consequences in order to bind the locally rational choice. A collective-action framing produces the same outcome when individually optimal actions aggregate to a suboptimal field-level result; the prescription’s landscape-binding architecture is the standard collective-action remedy applied to the optimization landscape itself

rather than to the actors operating within it. The prescription is not specified against the prisoner's-dilemma framing alone; it is specified against the field-level coordination failure all three structures jointly describe. The prisoner's-dilemma framing is adopted as the cleanest exposition of the marginal choice the paper isolates, not as the unique strategic structure the prescription is designed to bind. Whether the prescription survives substitution of strategic-structure assumptions across the alternatives in the cited convergence is the structural-robustness claim made here; this claim is itself open to disconfirmation if a strategic-structure alternative could be specified that the prescription's six properties do not bind, and Section 7 specifies the falsification regime under which such a counterexample would be evaluated.

The joint condition.

The three properties together produce the prisoner's-dilemma structure at the scale relevant to this paper. Each property in isolation would admit different interventions. The cost differential alone, addressed at the institution level, is what ceremonial governance interventions attempt; the within-institution remedy would be sufficient if the cost differential were the only structural property in play. The cost-of-stopping asymmetry alone could in principle be addressed by coordinated voluntary cooperation, if the path-dependent foreclosure did not progressively narrow the window during which voluntary coordination remained tractable. The path-dependent foreclosure alone could be tolerated if the cost differential and the cost-of-stopping asymmetry did not jointly drive the architectural commitments that produce the foreclosure.

The field-scale prisoner's dilemma emerges from the joint operation of all three. The cost differential establishes the human as a target of optimization. The cost-of-stopping asymmetry forecloses the voluntary-coordination class of remedies. The path-dependent foreclosure ensures that the window during which any remedy remains structurally available is closing while the trajectory continues to operate. The equilibrium under these three properties is human removal — not because anyone intends it, not because any individual actor benefits from it, and not because any single institution's governance has failed. It is the equilibrium because the three properties jointly determine what competitive optimization selects for under the present incentive architecture, and what it selects for is the trajectory.

This is the structural specificity that distinguishes the field-scale claim from generic-displacement claims and from domain-local failure-mode diagnoses. The three properties are not three separable problems to which separable solutions could be applied. They are the joint structure that makes the trajectory the default attractor of the field as a whole, and they require an architectural intervention that addresses the joint structure rather than its individual properties. Section 6 specifies what that intervention must look like.

Equilibrium without intent; restraint without binding.

No participant needs to intend the equilibrium. *Ordinary unilateral restraint* does not by itself dissolve the structure: an actor that merely exits continuation while leaving the surrounding optimization landscape unchanged usually removes itself from the competition without changing the payoff conditions governing those who remain. This is not a claim that no actor has asymmetric structural leverage. A regulator, compute provider, procurement authority, standards body, treaty bloc, insurer, or dominant platform operator may change the landscape if its action binds the conditions under which others operate. The point is narrower: unilateral restraint that does not bind the optimization landscape is structurally insufficient to redirect the equilibrium, even when undertaken by actors with otherwise substantial position. Landscape-binding action by an actor with the structural leverage to alter the conditions under which others operate is a different category of intervention; it is one of the routes by which the design

requirement of Section 6 could be implemented at the regulatory level, and it is part of the scope of the prescriptive contribution.

3.2 The Continuation Rationalization

Within this structural environment, individual actors construct rationalizations for continued participation that are locally coherent. The structurally predicted cognitive output of bounded rationality under the competitive conditions Section 3.1 specifies is what may be called the *continuation rationalization* or, in the rhetorical form it often takes, the *benevolent emperor* rationalization: *I cannot stop, because if I stop, someone less responsible will continue. Therefore the responsible course of action is for me to continue, more rapidly, so that the destination is reached by someone with values approximating mine rather than by someone whose values do not.*

The rationalization is structurally significant for two reasons.

First, it is the predictable cognitive output of bounded rationality under competitive pressure. The actor cannot fully model the long-term implications of their decision because the long-term implications are diffuse, probabilistic, and contingent on the actions of other actors whose decisions are themselves contingent. Faced with this irreducible uncertainty, the actor optimizes against what they can model — local competitive position, the perceived character of competitors, the actor's own intentions — and reaches a conclusion that justifies continuation. The conclusion is not corrupt. Under bounded-rational decision-making in conditions of irreducible uncertainty, rationalization of locally available reasoning is the dominant operating mode rather than an aberration. The prisoner's dilemma structure ensures that rationalization at the individual-actor level produces the trajectory regardless of whether the rationalization is correct in any particular case.

Second, the rationalization cannot be simultaneously decisive for all actors who invoke it, because each actor’s justification depends on a comparative premise — that competitors are worse stewards — that cannot hold for everyone at once. The rationalization may be correct for some actor along some dimension; it cannot ground the continuation of every actor who appeals to it. Structurally, however, this asymmetry does not block the trajectory. The destination is not determined by which actor reaches it first; it is determined by what the destination is, and the destination is the same regardless of who arrives there. The rationalization functions as the cognitive vehicle by which bounded rationality under competitive pressure produces continued participation in a prisoner’s dilemma whose equilibrium, under the present conditions, is human removal. Naming the rationalization does not impugn the actors who hold it. It identifies why each actor’s local reasoning produces a global outcome no actor would choose if presented with it directly.

The rationalization has access to a real reply that the foregoing argument should address directly: the destination might not be invariant under permutation of which actor arrives there, and architectural commitments made en route — what specifically gets built, who controls it, what values are baked in — are not identical across arriver-identities. The reply is correct as stated, and the response operates at a different level. The trajectory the paper names is evaluated against the human-non-removability standard specified in Section 2: substantive human contribution preserved across consequence-present systems where the system’s authority to operate depends on human judgment, accountability, legitimacy, or stewardship. The structural shape of the destination, evaluated against that standard, is invariant under permutation of arriver-identity — the human-non-removability condition is either met or not met, and the architectural commitments that produce the trajectory either preserve substantive human contribution or do not. Variations in what specifically gets built en route are real but

downstream of the standard the paper's evaluative reference identifies. An actor whose rationalization is "I will arrive faster so the destination is reached with values approximating mine" is making a claim about the values embedded in the destination, not about whether the destination preserves substantive human contribution. The values-embedded claim may be correct for some actor along some dimension; it does not address the standard against which the trajectory is being evaluated. The trajectory the paper names is the one in which substantive human contribution is removed, regardless of which actor's values are embedded in the systems that replace it. The rationalization's reply changes which destination-variant gets built; it does not change whether the destination satisfies the human-non-removability standard the paper takes as its evaluative reference.

3.3 The Trajectory Is Structurally Extrapolated From Documented Dynamics

The trajectory toward human removal is not presented here as an empirically validated forecast. It is a structural extrapolation from documented dynamics already operative in deployed AI systems. The paper's claim is not that the terminus has been observed, or that future capability can be predicted with certainty. The claim is that the documented dynamics identify the direction of the optimization gradient under the present incentive architecture, and that absent binding counter-architecture, continuation of those dynamics selects for progressive removal of substantive human involvement. The empirical record across high-consequence domains supports the extrapolation along multiple independent lines.

In high-consequence professional domains, clinical commentary has raised concern about the systematic substitution of AI-generated analysis for direct expert engagement, including the risk that it may produce degradation of expert capacity to identify cases requiring deviation from AI recommendations (Ferguson, 2025), and the Synthience corpus develops this as the calibration-drift mechanism (Gantz, 2026b). Recent empirical work confirms the concern. In a multicentre observational study of endoscopist performance, the standard non-AI adenoma detection rate fell measurably after exposure to AI-assisted colonoscopy, with the rate dropping from 28.4% (226/795) before exposure to 22.4% (145/648) after (Budzyń et al., 2025). The mechanism is calibration drift: practitioners who do not exercise full diagnostic reasoning lose access to the reference standard against which AI recommendations would need to be evaluated. The result is not a stable equilibrium in which the practitioner maintains oversight while the AI handles routine cases. It is a progression in which the boundary between "routine" and "non-routine" itself shifts as the practitioner's capacity to recognize non-routine cases erodes.

The pattern recurs across consequence-present domains in which AI advisory systems are deployed alongside expert human judgment. In mammography, incorrect AI suggestions impair radiologist performance, with effects modulated by the format and salience of the AI output (Dratsch et al., 2023). A pilot study of explainability and attitudinal-priming interventions in mammography found that radiologist consultation rates and accuracy under incorrect AI assistance shifted substantially, with accuracy degrading sharply when AI suggestions were wrong (Mehrizi et al., 2023). In computational pathology, automation bias has been documented in 7% of cases among 28 pathology experts, with time pressure increasing the severity of the bias and shaping the conditions under which experts defer to incorrect machine output (Rosbach et al., 2026). In algorithmic decision-support deployed in judicial bail proceedings, approximately 90% of judges who override the algorithmic recommendation underperform the algorithm on those overrides, and roughly 70% of override decisions perform no better than random (Angelova, Dobbie, and Yang, 2025); a foundational study of hiring discretion documents a structurally analogous pattern in which managerial discretion against test-based recommendations systematically degrades outcomes (Hoffman, Kahn, and Li, 2018). The

aggregate pattern is consistent across these studies: where human judgment is preserved formally alongside AI advisory output, the substantive contribution of human judgment can be eroded, inverted, or rendered indistinguishable from random under the conditions in which the systems actually operate.

The displacement of expert judgment by AI-mediated advisory systems is a special case of the broader pattern. The lineage of the more general structural diagnosis runs through institutional sociology. Meyer and Rowan (1977) documented the decoupling of formal organizational structure from substantive work, identifying ceremonial conformity as the predictable equilibrium under conditions where the organization derives legitimacy from formal compliance independent of substantive function. Subsequent work has extended the diagnosis to compliance regimes specifically, showing how decoupling between compliance and substantive practice produces the failure pattern even where compliance is formally rigorous (Schembera, Haack, and Scherer, 2023). In organizational governance around AI deployment, ceremonial review structures systematically converge on documentation rather than challenge, because the cost of challenge exceeds the cost of approval and bounded rationality under cognitive overload predictably selects the path of least resistance (Gantz, 2026d). The Boeing 737 MAX trajectory, the Theranos trajectory, and the financial governance failures across multiple regulatory cycles are each documented cases in which the human governance role was structurally present but was not actually exercising the function the architecture was designed to ensure. The pattern is not specific to AI. AI accelerates it because it lowers the cost of producing documentation that satisfies the formal requirements of governance without producing the substantive engagement governance was designed to require.

In labor and organizational structure, the displacement of human cognitive labor by AI-mediated alternatives has moved from prediction to ongoing process. Automation pressure has produced documented displacement effects in labor markets through robotic substitution in manufacturing and related sectors (Acemoglu and Restrepo, 2020); the argument here extends that documented substitution logic to AI-mediated cognitive work where the capability margin makes substitution locally rational. The extension is structural rather than directly established by the labor-economics citation alone. The displacement does not require the AI to fully replicate the displaced cognitive function. It requires only that the AI-mediated alternative be sufficiently close to the function for the cost differential to drive substitution at the margin, and the margin moves with each improvement in capability.

Each of these documented dynamics is a fragment of the trajectory. The trajectory is not a separate phenomenon from the documented patterns; it is the structural extrapolation of those patterns under the same incentive conditions that produced them, applied across the time horizons in which AI capability continues to develop. The extrapolation does not require a specific forecast of full human-function replication or a dated capability threshold. It requires only that the documented dynamics continue to operate, that the incentive structure driving them continues to be the dominant force shaping development decisions, and that the capability-margin dynamic specified in the closing paragraph of this section continues to cross substitution thresholds in consequential domains.

The relationship between the cited evidence and the trajectory claim should be made explicit, because the cited studies document a different finding than the trajectory itself. The studies document local performance effects within human-AI dyads — deskilling after AI exposure, automation bias under incorrect AI suggestions, override-underperformance in algorithmic decision-support — under conditions in which humans remain formally present in the loop. The studies do not document an optimization gradient toward removing humans from the loop. The trajectory claim is what the structural argument adds on top of the literature.

The structural argument operates in three stages. The first stage is the observed pattern: under competitive pressure, human contribution within the loop becomes substantively reducible, with the substantive contribution eroded, inverted, or rendered indistinguishable from random under the conditions in which the systems actually operate. This is what the cited literature documents directly.

The second stage is the transition the literature also documents under the description "human structurally present, substantively absent" (Section 5.2): once substantive contribution is reducible, the human position within the loop becomes ceremonial residue — formally preserved, materially inactive — across the conditions where the substantive reduction has occurred. This stage is documented in the same literature, framed as the gap between formal presence and substantive contribution.

The third stage is the structural extrapolation the paper adds: ceremonial residue itself is selected for removal where the cost of formal preservation exceeds the cost of trust, legitimacy, regulatory, or political consequences of visible removal, and Property 1 of Section 3.1 selects for that removal at the margin where the cost relation favors it. Section 5.3 develops the structural argument for why the ceremonial residue does not constitute a stable terminus; the optimization gradient that produced the substantive transfer continues to operate on the formal residue under the same logic. The first two stages are what the empirical record shows. The third stage is what the structural argument predicts under the present incentive conditions, and is the stage at which Section 5.3's destabilization argument is load-bearing for the trajectory's continuation past visible ceremonial preservation.

The cited literature is compatible with an alternative reading: that human-AI dyads need better design, and that better-designed dyads would preserve substantive human contribution without erosion. This reading does not block the trajectory claim. "Better design of human-AI dyads" is precisely what the design requirement of Section 6 specifies — the architecture that binds the optimization landscape so that substantive human engagement is the path of least resistance. Absent the design requirement being satisfied, the structural prediction is that the optimization gradient does not stop at the first-stage degradation pattern the literature documents; it continues through the second-stage ceremonial residue the literature also documents and onward to the third-stage architectural removal the structural argument extrapolates. The literature supports the in-loop-degradation finding either way; whether the trajectory continues through its later stages depends on whether the architectural conditions specified in Section 6 are satisfied or whether the gradient continues to operate against substantive human contribution. A literature finding that in-loop degradation reverses durably without architectural changes satisfying Section 6.1 would be a clean disconfirmation of the structural prediction, in the sense Section 7 specifies.

The trajectory is not merely one imagined scenario among unrelated possibilities. It is the default structural attractor identified by the documented dynamics when those dynamics are interpreted under the present incentive conditions, absent binding counter-architecture.

The argument does not require a forecast that AI systems will fully replicate human cognition, nor does it require a claim that any specific capability threshold will be reached by a specific date. It does, however, depend on the continuation of the capability-margin dynamic already visible in deployed systems: as AI-mediated alternatives become sufficiently close to particular human functions, substitution becomes locally rational at that margin. The trajectory therefore depends not on speculative parity with the full human function, but on repeated local crossings of substitution thresholds under the same incentive conditions. If capability improvement stalled before such crossings reached consequential domains beyond those already documented, the trajectory would be correspondingly limited. That limitation is part of the claim's conditional

structure: the structural argument is conditional on the capability-margin dynamic continuing to operate, and on the incentive architecture that selects for substitution at the margin continuing to be the dominant force shaping development decisions. Both conditions are presently obtaining and have been obtaining throughout the period the empirical citations document. Whether they will continue to obtain across the longer horizon is part of what the disconfirmation criteria in Section 7 specify as testable.

Whether the actors making the optimization decisions face structural displacement themselves under sustained AI development is an empirically open question this paper does not address. The argument that follows concerns the trajectory of human displacement under competitive optimization pressure and the governance architecture that could change it.

4. Counterforces: Why the Trajectory Is Not Self-Correcting

The trajectory does not run unopposed. Counterforces exist: regulatory action, customer preference, professional licensing, technical limits, public concern, internal company resistance, labor organization, reputational pressure, and the open-source and public-interest research communities. The question this paper engages is not whether such counterforces exist but whether they bind the optimization landscape such that the trajectory is reversed rather than merely slowed or locally moderated. The structural claim of this section is that counterforces operating at any single level, without architectural support that binds the optimization landscape itself, are predictably routed around under sustained competitive pressure. They may delay the trajectory; they do not redirect it. The argument proceeds counterforce by counterforce.

Regulation as currently practiced does not bind the optimization landscape. Voluntary frameworks fail structurally. An empirical audit of voluntary AI commitments to the White House found an average compliance score of 52% across the 16 companies that signed, with 11 of 16 companies scoring 0% on model-weight security commitments (Wang et al., 2025). The structural diagnosis of why voluntary regulatory adoption fails — the absence of enforcement architecture that binds the optimization landscape rather than asserting a preference — has been developed independently (Kingsbury Barry, 2026a). Mandatory regulation faces lobbying pressure from the entities most needing to be governed; in the most consequential AI deployment sectors, the regulatory environment is not currently trending toward binding constraint. Recent analysis of AI regulation identifies systematic gaps in oversight of internal deployment, including scope ambiguity, point-in-time compliance frames, and information asymmetries that the regulated entities exploit (Kwon and Casper, 2026). Regulation that documents compliance without binding the gradient that opposes compliance is precisely the ceremonial governance failure mode documented across high-consequence domains (Meyer and Rowan, 1977; Schembera, Haack, and Scherer, 2023; Gantz, 2026d, 2026e).

Customer preference does not bind the optimization landscape. A preference for human-mediated services exists in many domains and is sometimes expressed at the point of consumption. But the relevant consequential AI deployment is not, in most cases, in domains where customers select architecture. In healthcare, finance, employment, criminal justice, and most administrative interactions with consequential institutions, the customer is the recipient of architecture chosen by others. Customer preference operates within consumption choices that do not control the architecture of the consequential systems they encounter. Where customer preference does control architecture, it operates against price differentials that, under the cost gradient described in Section 3.1, systematically disfavor the preserved-human option.

Professional licensing does not bind the optimization landscape at the relevant level when it operates as individual-practitioner licensing alone. Licensing operates on individual

practitioners, not on the architectural decisions about how AI is deployed in the institutions in which those practitioners operate. A licensed physician working within an institutional system that has architecturally minimized the practitioner's capacity to deviate from AI-mediated recommendations is structurally constrained even if the license formally authorizes deviation. The licensing constraint is preserved; the substantive function it was designed to license is eroded by the surrounding architecture. The ceremonial-substantive gap (Meyer and Rowan, 1977; Schembera et al., 2023) operates inside the licensed role.

The argument applies to professional licensing as currently constituted in most domains, in which licensing addresses the practitioner's authorization to perform the function rather than the institutional architecture surrounding the function. *Licensing-as-institutional-governance-architecture* — in which the licensing instrument is structured to bind not only the practitioner but the institutional conditions under which the practitioner's substantive engagement remains operationally tractable — is a different intervention. SI-WP-008 specifies one such architecture for the medical case, in which competence-linked credentialing tied to demonstrated AI oversight capability operates alongside institutional accountability for drift patterns (Gantz, 2026e). Such licensing-architecture interventions can bind under the design requirement of Section 6 if they are specified to satisfy the path-of-least-resistance condition. The distinction matters: professional licensing as a counterforce that operates only on the individual practitioner does not bind the trajectory; licensing reconfigured as institutional governance architecture, properly designed, can.

Technical limits do not bind the trajectory in the way the framing implies. The trajectory does not depend on AI capability reaching some threshold of full replication of the displaced human function. It depends on competitive pressure to substitute AI for humans wherever the substitution is locally rational at the present capability margin. The margin moves with each capability improvement; the pressure does not require the margin to be at parity with full human function. As the empirical record of automation bias and override failure documents (Section 3.3), the substitution proceeds at margins well short of capability parity, because the cost differential is sufficient to drive substitution before the capability differential is closed.

Reputational pressure and internal resistance moderate locally but do not bind globally. Reputational pressure can produce visible reversals in specific cases. But the reversals operate at the level of public-facing decisions; they do not modify the underlying optimization gradient. Internal resistance from employees can locally slow specific deployments but is itself subject to the competitive pressure that produces the trajectory: organizations that allow internal resistance to substantially slow deployment lose ground to organizations that do not. The structural feature of restraint — that its costs are immediate and its benefits are diffuse — applies to internal resistance as much as to external regulation.

Public concern does not bind the optimization landscape unless converted into enforceable architecture. Public concern can alter reputational cost, electoral pressure, customer behavior, and regulatory attention. It is therefore not nothing. But concern as concern remains external to the deployment architecture. It can create moments of salience, and those moments can matter. They redirect the trajectory only if translated into constraints that change the cost of removal relative to preservation. Without that translation, public concern follows the same pattern as reputational pressure: visible local moderation without durable field-level binding. The window during which public concern produces translation into binding architecture is narrower than the window during which the underlying concern persists, because translation requires regulatory or institutional architecture that satisfies the design requirement of Section 6, not merely the demand that something be done.

Labor organization can contest the trajectory but does not by itself bind the field-level gradient. Labor organization can resist displacement, preserve bargaining power, and force institutional concessions. It is a real counterforce, not merely symbolic. But labor organization operates unevenly across domains, jurisdictions, and worker classes, and it is weakest precisely where AI-mediated substitution is most easily externalized, deskilled, or moved across institutional boundaries. It can bind locally where it controls deployment conditions; it cannot redirect the field-level trajectory unless its constraints become embedded in organizational or regulatory architecture satisfying the design requirement. The historical pattern of labor organization successfully constraining displacement under sustained automation pressure is sectoral and conditional; the structural prediction is that it will operate similarly against AI-mediated displacement: locally significant where it is structurally supported by binding architecture, locally insufficient where it is not.

Open-source and public-interest research communities alter capability distribution but do not automatically preserve human-non-removability. Public-interest research can expose risks, produce safer tools, develop audit methods, and contest proprietary concentration. Open-source communities can democratize access and reduce dependence on closed actors. But neither openness nor public-interest orientation by itself binds the optimization landscape against human removal. In some cases, broader capability diffusion may intensify the cost-of-stopping asymmetry by increasing the number of actors able to continue. These communities become trajectory-altering only when their outputs support enforceable, non-routable, path-of-least-resistance governance architecture. The contribution of public-interest research to the eventual production of such architecture is real and may be substantial; the production of such architecture is not the automatic consequence of the contribution.

The confidence trap is a meta-counterforce that fails in a structurally distinctive way. Long periods without visible catastrophe in deployed AI systems will be interpreted as evidence that current governance is sufficient, generating self-reinforcing under-investment in correction even as latent risk accumulates. This pattern has been formalized in operations management as the *confidence trap* mechanism, which produces self-reinforcing belief that controls are working precisely during the periods in which the controls are accumulating the conditions for their own failure (Bhardwaj and Akkermans, 2024).

The cases this paper cites alongside the confidence-trap framing — Boeing 737 MAX, Theranos, financial governance failures — eventually produced visible catastrophes that triggered reassessment. The Boeing trajectory ended in two crashes concentrated in time and attributable to a specific architectural decision; the Theranos trajectory ended in laboratory failures concentrated enough to attribute and prosecute; the financial-governance trajectories ended in market collapses concentrated enough to force regulatory response. In each case, the trajectory was gradual but the failure mode that triggered correction was discrete and visible. The confidence trap operated until the catastrophe; the catastrophe produced enough concentrated harm to overcome the confidence trap.

For the trajectory toward human removal in advanced AI deployment, the failure mode is structurally different. The harms produced by AI-mediated decision degradation are statistically distributed across populations rather than concentrated in discrete events. A measurable rise in adenoma miss rates across endoscopist populations does not produce a Lion Air crash. A documented automation-bias rate in pathology does not produce a Theranos. A 90% override-underperformance pattern in judicial bail decisions does not produce a 2008-style market collapse. The harms are real and aggregate to substantial cost across populations, but the cost is distributed thinly enough across cases, time, and institutions that no single event meets the salience threshold at which reassessment is forced.

Distributional-harm detection infrastructure is not absent in these domains: cancer registries, adverse-event reporting systems, epidemiological surveillance, judicial-outcome tracking, and the academic research that produced the Budzyń, Rosbach, Angelova, Hoffman, Mehrizi, and Dratsch citations are products of exactly this infrastructure, and the cited evidence in this paper is itself the output of distributional-harm detection. The argument is therefore not that detection is absent; it is that detection alone does not constitute correction. Detection produces evidence; evidence becomes correction only when it translates into binding architecture satisfying the design requirement of Section 6. The historical concentrated-catastrophe cases produced corrective architecture only after events forced political-salience translation: the regulatory and institutional changes that followed Boeing, Theranos, and the financial collapses were responses to events that overwhelmed the confidence trap, not to distributional-harm reports that preceded the events. The structural prediction is that AI-mediated distributional harm produces evidence at lower salience thresholds than concentrated catastrophe, that the evidence accumulates within the documenting infrastructure (publications, registries, audits) without forcing the translation into binding architecture, and that whether translation occurs follows the same counterforce pattern that operates across the other counterforces this section documents — locally significant where structurally supported by binding architecture, locally insufficient where it is not. The confidence trap operating against distributed degradation is therefore not the same dynamic as the confidence trap operating against trajectories that eventually concentrate. The longer the trajectory operates, the more confidence accumulates; the structure of the harm does not produce the events that would otherwise interrupt the accumulation, and the detection infrastructure that does operate produces evidence that does not by itself force architectural translation.

The asymmetry between the historical cases and the AI trajectory is structurally deeper than distribution-versus-concentration alone. The Boeing, Theranos, and financial-governance trajectories ran on optimization gradients that did not select for catastrophe-invisibility. Boeing was optimizing for cost reduction; the two crashes were the unintended consequence of a gradient that did not select for crash-invisibility. Theranos was optimizing for capital and credibility; the laboratory failures were the unintended consequence of a gradient that did not select for failure-invisibility. The financial-governance trajectories were optimizing for return; the collapses were the unintended consequence of gradients that did not select for collapse-invisibility. In each historical case, the corrective catastrophe emerged because the gradient did not actively suppress the conditions for its emergence.

The AI-mediated trajectory differs in kind: its harm pattern is statistically distributed, lagged, individually small, and confounded by exogenous factors not because of contingent features of the deployment context, but because the optimization gradient producing the trajectory selects for those features. AI-mediated decision support distributes errors across populations because population-scale processing is what produces the cost differential Property 1 identifies; the lagging, individually-small, confounded character of the harms is downstream of the substitution gradient, not a contingent property of the harm. The confidence trap operating against the AI trajectory is therefore operating in a regime the gradient actively maintains. The corrective catastrophe-event that overcame confidence trap in the historical cases is precisely the form of event the AI-mediated gradient is structurally biased against producing, because the form of harm the gradient produces is the form that does not concentrate.

This is an empirical claim about the rate and concentration of catastrophe production under AI-mediated systems on the timescales at which architectural decision points close, not a structural certainty about the trajectory's eventual failure mode. It is in principle falsifiable: a discretely catastrophic AI deployment failure that triggered field-level reassessment would constitute evidence against this argument's claim about the failure mode's distributional character on the

relevant timescale. But the structural prediction is that the trajectory's failure mode operates distributionally on the timescales at which the architectural decision points are closing, that confidence-trap accumulation therefore continues without the corrective events that operated in the cited historical cases, and that the absence of visible AI deployment catastrophes during this window is not evidence of governance adequacy under this trajectory's specific dynamics. Whether the trajectory eventually produces concentrated catastrophe of the kind that overcame confidence trap in the historical cases is a separate question not load-bearing for the present argument; by the time such concentration occurred — if it did — the path-dependent foreclosure of Section 3.1 Property 3 would have substantially narrowed the architectural decision space the prescription depends on.

The structural conclusion. Counterforces operating at any single level fail to bind because the optimization landscape rewards routing around them. The empirical record across high-consequence domains is consistent: ceremonial governance, voluntary frameworks, preference-stating regulation, isolated professional licensing, and internal resistance produce local moderation under specific conditions and predictable erosion under sustained competitive pressure. The structural prediction is that, under sustained competitive pressure, non-architectural counterforces — whether operating singly or in combination — will tend to be eroded along the patterns documented here, because none alters the optimization landscape's selection of the locally rational path. The counterforces examined in this section cover the dominant non-architectural counterforce types presently advocated in AI governance discourse, and the historical record across high-consequence domains repeatedly shows ceremonial governance, voluntary frameworks, and unbound professional norms failing when they attempt to hold against an optimization gradient. Whether a novel non-architectural counterforce, or a novel combination of softer measures, could falsify this prediction is an open question; the design requirement of Section 6 identifies the architectural properties that would make binding robust, should such a counterforce or combination not emerge. Producing a non-architectural counterforce that successfully binds the optimization landscape under the conditions Section 3 specifies would falsify the prediction. The argument of this section is not that counterforces are pointless. It is that counterforces operating without architectural support are subject to the path-of-least-resistance dynamics this section documents, and that the time spent expecting them to redirect the trajectory is time during which the architectural decision points continue to close.

5. The Destination

This section names the destination structurally without specifying the operational architecture by which it is reached. Implementation-level specifications are outside the scope of this paper for the reasons stated in the methodological positioning above. What follows is the destination as a structural condition, not as an implementation roadmap.

5.1 The Terminus of the Optimization Gradient

The optimization gradient does not contain, by itself, an interior stopping point that preserves substantive human contribution as such. The pressure to reduce dependence on the human variable continues until the dependence is structurally minimized — meaning, until the human is removed from every loop in which the human's presence is a cost that an AI-mediated alternative can absorb. The gradient may leave humans in place where human presence remains useful for trust, legitimacy, liability, regulatory standing, customer willingness, political necessity, or strategic differentiation. Those are real conditions and they may produce durable retention in specific domains. But they are contingent stopping points generated by surrounding constraints or advantages, not by the gradient's preservation of the human function as such. Under the present incentive conditions, the gradient stops preserving the human wherever the

human contribution ceases to be cost-protective, function-constitutive, legitimacy-constitutive, or externally required. The gradient does not stop when the human's presence is reduced. It stops when the human's presence is no longer the marginal cost being optimized, or when surrounding architecture makes preservation of the human contribution structurally cheaper than removal.

The terminus is not "AI replaces humans in some tasks." That description applies to the present and to most points along the trajectory. The terminus is the structural condition in which the human's continued presence in any consequential loop is preserved only by structural commitments that bind the optimization pressure rather than by the human's continued utility within the loop. At the terminus, the human is in the loop only where governance architecture has made removal costlier than continuation. Everywhere else, the human is gone.

This is a structural condition, not a science-fictional scenario. It does not require AI systems to develop motivations of their own. It does not require any single actor to intend it. It requires only that competitive optimization under the present incentive structure continues to operate on the timescales required for the trajectory to complete. The completion is structurally determined under those conditions; intervention that binds the conditions can redirect it.

5.2 What "Human Removal from the Relational Loop" Means

The phrase requires precision. Per the definitions in Section 2, *removal from the relational loop* does not mean the elimination of the species, a singular catastrophic event, or the absence of human bodies from formal organizational charts. It means the structural condition in which the substantive human contribution to consequential decisions, system operation, and institutional governance has been reduced below the threshold at which the human's presence shapes the outcome.

A human can be physically present, formally credentialed, structurally positioned, and visibly involved in a process while contributing nothing to the substantive output of that process. The empirical record across high-consequence domains documents this pattern at scale: the erosion of detection rates among practitioners exposed to AI-assisted diagnosis (Budzyń et al., 2025); the documented automation-bias rates in pathology and the impairment of radiologist performance under incorrect AI suggestion (Rosbach et al., 2026; Dratsch et al., 2023; Mehrizi et al., 2023); the override-underperformance pattern in algorithmic recommendation across judicial bail and managerial hiring (Angelova, Dobbie, and Yang, 2025; Hoffman, Kahn, and Li, 2018); and the bounded-rationality dynamics described in the Synthience corpus (Gantz, 2026d). The human in each case is structurally present and substantively absent. The loop has already been redefined to exclude the function the human's role was originally designed to provide.

The terminus is the generalization of this pattern across the loops where it has not yet completed. The human becomes the formal residue of a function that has been substantively transferred. The architecture continues to require human presence only because human presence is required by the architecture; the architecture no longer requires human presence because the human's presence accomplishes anything.

For the purposes of this paper, *arrival* at the destination does not require the disappearance of human bodies from institutions, the abolition of human-facing procedures, or a single terminal event. The destination is reached within a domain when three conditions jointly obtain. *First*, consequential outputs are generated by AI-mediated or AI-dependent processes whose operation no longer depends on substantive human judgment. *Second*, human participation, where preserved, no longer materially changes outcomes under realistic operating conditions.

Third, restoration of substantive human contribution would require architectural reconstruction rather than ordinary procedural correction. Arrival is the durable joint obtaining of these three conditions, on the same durability standard Section 7's disconfirmation criteria specify: persistence under sustained competitive pressure, not transient instantiation under narrow circumstances. A domain in which the three conditions obtain briefly under specific market or regulatory conditions but recede under perturbation has not arrived in the structural sense the trajectory claim names; arrival is the condition the gradient maintains against perturbation, consistent with the attractor framing in Section 2. The field-level destination is the generalization of this durable condition across consequence-present domains. Each of the three arrival conditions is observable in principle, and each follows from the Section 2 definition of substantive human involvement. The arrival criteria are stated here so that the destination claim has a structural test rather than only a rhetorical specification.

5.3 The Destination Is Not Stable

A condition is sometimes treated as if its stability were a feature: once reached, it persists. This is incorrect for the destination described here. The condition in which the human is the formal residue of a function that has been substantively transferred is not a stable terminus. It is a transitional condition under sustained optimization pressure.

The optimization pressure that produced the substantive transfer continues to operate after the transfer is complete. The formal residue of human presence becomes itself a cost that the optimization pressure works to minimize, because the formal residue requires architectural accommodation, regulatory deference, legitimation infrastructure, and maintenance of the systems that preserve it. Each of these is a cost that the optimization gradient identifies as removable under the conditions producing the gradient.

The progression from "human as formal residue" to "human as removed" is the same gradient that produced the previous progression, operating on the same logic. There is no structural feature of "formal residue" that resists the gradient. There is only the absence of architectural commitment binding the gradient against the residue, and that absence is precisely what produced the previous progression in the first place.

The destination, in the structural sense the paper names, is the condition in which the optimization gradient has worked through the formal residue to its terminus. What that condition means for human authority-bearing participation in consequential systems, beyond the structural destination specified here, is not elaborated within this paper's scope. The argument of this paper does not require that specification. It requires only the recognition that the gradient does not stop where the formal residue begins, and that the formal residue is therefore not the resting point it is sometimes treated as.

The destabilization argument of this subsection requires one further clarification, because the same logic could appear to apply to the prescribed architecture of Section 6 once implemented. If no formal feature resists the gradient, would the gradient not also destabilize the binding architecture once it is in place? The answer turns on a structural distinction between the residual-human condition this subsection destabilizes and the prescribed architecture Section 6 specifies. The residual-human condition is unstable because the human is structurally cost-bearing while the gradient continues to operate against the cost: the gradient minimizes cost, the human is cost, the formal residue is therefore subject to continued minimization. The prescribed architecture inverts this relationship through the three mechanisms Section 6.1.1 specifies. Removal of substantive human contribution degrades the function rather than reducing the cost of producing the function (function-constitutive dependence); bypass of substantive human contribution is operationally costlier than honoring it (substrate cost-

inversion); non-compliance with the human-non-removability requirement loses access to operational conditions like procurement, certification, and market access (regulatory externalization). Under those conditions, the gradient operates *for* the architecture's preservation aim rather than against it. The architecture is not a binding-against-the-gradient — which Section 5.3 correctly identifies as unstable — but a redirection-of-the-gradient, in which the same optimization pressure that otherwise drove removal now drives compliance because compliance is the structurally cheaper path. This is what the design requirement specifies and what the cost-inversion subsection of Section 6 develops in detail.

The load-bearing distinction can be stated cleanly: the residual-human condition is unstable because the gradient operates against it; the prescribed architecture is stable in the relocated sense because the gradient is redirected to operate for it. That distinction is what the prescription rests on, and it is the structural feature that prevents the destabilization argument of this subsection from applying to the prescribed architecture in the same form it applies to the residual-human condition.

The prescription does not solve the stability problem this subsection identifies; it relocates it. The residual-human instability is replaced by a regulatory-coordination instability: the architecture’s stability depends on the three mechanisms continuing to hold, and the regulatory layer in particular depends on the binding-coalition conditions Section 6.3 leaves open. The same prisoner’s-dilemma optimization that produces the field-level trajectory operates on regulators competing across jurisdictions, on procurement coalitions facing fragmentation pressure, and on standards bodies subject to capture; the architecture has no internal mechanism for making the regulatory layer self-stabilizing once it is in place. Whether the stability problem in its relocated form is structurally easier than its substrate-level form is the open question the prescription’s long-run viability turns on. The substrate-level form is unstable by Section 5.3’s argument; the regulatory-coordination form is uncertain in the sense Section 6.3.2 specifies — the regulatory layer faces the same prisoner’s-dilemma optimization the field-level diagnosis identifies for AI developers, and whether coordination success against that pressure is possible is the bootstrap problem the prescription’s long-run viability turns on, not a separate stability claim with its own evidentiary status. The prescription is a stability-relocation rather than a stability-resolution, and the relocated problem is the open question Section 6.3.2 develops further.

Whether the relocated stability problem is structurally easier than the substrate-level form is open in the strict sense — the paper does not resolve it — but the relocation is not arbitrary. The substrate-level form has no historical precedents in which the gradient operating against a structurally cost-bearing target has been bound durably under sustained competitive pressure; the residual-human condition Section 5.3 destabilizes is precisely the configuration in which no such precedent exists. The regulatory-coordination form has partial precedents: financial regulation under deposit-insurance and central-banking architecture has bound competitive pressure across jurisdictions for extended periods despite continuous capture pressure and arbitrage incentives; aviation safety certification has bound an active counterparty (manufacturers) across jurisdictions through architectures involving the four forcing-function conditions Kingsbury Barry (2026b) specifies in partial form; pharmaceutical pre-market review has bound an active counterparty for narrow technical scope. None of these is a complete precedent for the AI governance case, and the disanalogies developed in Section 6.3 are real. The point of the partial precedents is narrower: the regulatory-coordination form has historical instances in which the relevant binding has held against optimization pressure for sustained periods, and the substrate-level form does not. This does not establish that the relocated problem is solvable; it establishes that the relocation moves the prescription into a structural class where partial precedents exist, rather than into a class where they do not. The asymmetric

precedent base is part of why the relocation is structurally meaningful even though it does not resolve the stability problem.

6. The Governance Architecture That Could Change the Outcome

This section specifies the governance architecture that could change the trajectory described in Sections 3, 4, and 5. The specification is structural, not operational. It states the design requirement that any intervention capable of redirecting the trajectory must satisfy, and identifies the levels at which the requirement must bind. The specification does not promise implementation. The claim is one of necessity and conditional structural adequacy, not implementation guarantee: any intervention capable of redirecting the trajectory must, at minimum, satisfy this design requirement (necessity), and interventions failing the requirement will predictably erode under the dynamics documented in Section 4; if the design requirement is satisfied at the relevant scope and persists under sustained competitive pressure, the mechanisms identified by the prescription should preserve human-non-removability against the specific dynamics this paper diagnoses (conditional structural adequacy). Whether such an intervention would actually be adopted, scaled, enforced, or preserved under all adversarial institutional conditions is a separate question this paper does not adjudicate.

The section proceeds in five steps. Section 6.1 states the design requirement and explains how the apparent cost contradiction with Section 3.1's diagnosis is resolved through gradient inversion. Section 6.2 documents why lesser interventions fail. Section 6.3 specifies the three architectural levels at which the requirement must bind, addresses the premature-architectural-commitment critique, and names the regulatory bootstrap problem the prescription's load-bearing layer creates. Section 6.4 demonstrates translatability across three consequence-present patterns. Section 6.5 specifies the closing window during which architectural intervention remains structurally available.

6.1 The Design Requirement

The design requirement is stated by Kingsbury Barry and Montanez (2026b): governance architecture must make compliance with the governance constraint the path of least resistance under the same incentive pressures that otherwise drive non-compliance. The requirement is stated as a property of the architecture, not as an aspiration of the actors operating within it.

For the specific governance constraint relevant to the trajectory this paper names — the constraint that human involvement in consequential AI-mediated systems must be structurally non-removable rather than merely preferred — the design requirement is as follows. The architecture must make the inclusion of substantive human involvement, at the loops where the trajectory toward removal otherwise operates, the path of least resistance under the optimization pressure that drives the removal. Compliance with the constraint must dominate non-compliance across the selection dimensions operative in the relevant deployment context — cost, speed, competitive advantage, and institutional tractability — such that any local advantage non-compliance retains on one dimension is outweighed, disabled, or externalized by the architecture under the same competitive conditions that presently make non-compliance the dominant strategy. These four dimensions are not independent further requirements layered on top of the path-of-least-resistance criterion; they are the surface area across which the criterion operates. An intervention satisfies the design requirement when compliance is the dominant path on the dimensions on which the optimization pressure operative in the relevant deployment context actually selects.

This is not a description of how to make actors want to comply. It is a description of how to bind the optimization landscape such that compliance is what the optimization landscape selects for. The actors' wants are structurally irrelevant to the design. The design changes the gradient, and the gradient produces compliance regardless of the actors' wants, in the same way that the present gradient produces non-compliance regardless of the actors' wants.

6.1.1 What It Means to Invert the Gradient

The design requirement of 6.1 names what compliance must look like under the optimization pressure that otherwise drives removal. It does not, by itself, explain how the cost relationship can be inverted. Property 1 in Section 3.1 establishes that human involvement is, under the present incentive architecture, a structurally favored target of marginal cost reduction in consequential AI deployment. Section 6.1 says compliance with the human-non-removability constraint must become cheaper, faster, more competitively advantageous, and more institutionally tractable than non-compliance under the same competitive conditions. These two claims sit in tension on their face. The work of this subsection is to specify how the architecture resolves the tension — how the cost-bearing structure can be changed without contradicting the diagnosis from which the prescription follows.

Making human-non-removability the path of least resistance does not mean making human involvement intrinsically cheaper than AI-mediated substitution in every local task. That would contradict the diagnostic argument. The design requirement instead operates by changing the cost-bearing structure of the system, through three structurally distinct mechanisms operating at the three levels Section 6.3 specifies.

The first mechanism operates at the individual level, by making the protected human contribution architecturally constitutive of the function the system is authorized to produce. Removal degrades the output rather than reducing the cost of producing it. The cost of preserving the human contribution does not change in absolute terms; what changes is what the system *is* when the contribution is removed. Consider a diagnostic system whose authorization to operate depends on substantive practitioner engagement and whose output is materially shaped by that engagement under realistic conditions: such a system does not save the cost of practitioner engagement when engagement is reduced. It produces a different system, with different authorization, at the cost of the function the original system was authorized to provide. The cost of removal is the loss of the function rather than the saving of the function's cost. The structural property — that the substantive human contribution is constitutive of the function the system is authorized to produce — sits in conversation with related work in adjacent literatures, including human-factors research on automation surprises and the ironies of automation, high-reliability organization theory's treatment of expert judgment as a structurally-required component of safety-critical operation, and the supervisory-control literature's analysis of human roles in automated systems. Those literatures share the analytical orientation that the human contribution is not a removable supplement to the system, but they articulate the property differently and at different scales than the architectural-property claim Section 6.1.1 is making. The Continuity Anchoring Method (Gantz, 2026a) specifies one architecture that produces the property in the form Section 6.1.1 requires; the property itself is the structural requirement, and other architectures meeting it would also satisfy the design requirement at this level.

The second mechanism operates at the organizational level, by structuring the institutional substrate so that honoring substantive human contribution is operationally cheaper than reconstructing, simulating, or routing around it. Canonical state, role authority, verification continuity, artifact lineage, and propagation constraints — the substrate components specified by the Operational Continuity Architecture (Gantz, 2026f) and the Institutional Continuity

Substrate (Gantz, 2026g) — can be designed so that the operational cost of substantive engagement is lower than the operational cost of bypass. The structural property — that institutional substrate can be designed to make substantive engagement operationally cheaper than its bypass — has analogs in adjacent literatures on distributed cognition (institutional artifacts as cognitive scaffolding whose disruption imposes coordination cost rather than saves it), coordination-cost theory in organizations, and organizational routines (substantive routines as path-dependent structures whose bypass requires reconstruction rather than substitution). Consider an organizational governance review function in which the substrate has been designed so that producing a documented evaluation that respects canonical institutional state is faster and operationally simpler than producing a parallel evaluation that ignores or simulates it: under those conditions, the bounded-rational reviewer optimizing against operational cost takes the path the substrate makes cheap, and the substrate has been designed to make that the path that produces substantive evaluation rather than ceremonial documentation. The cost of human contribution is not made artificially low; the cost of bypassing it is made structurally high. Under bounded-rationality conditions in which actors optimize against operational cost, the path of least resistance becomes the path that respects the substrate, because the substrate has been designed to make that path operationally easier.

The third mechanism operates at the regulatory level, by externalizing the cost of removal onto the architecture deploying the AI system. Non-compliant architectures can be made slower to deploy, more expensive to operate, legally unavailable, non-certifiable, non-procurable, non-insurable, or non-deployable under the forcing-function conditions Kingsbury Barry (2026b) specifies. Consider a procurement architecture in which government, enterprise, and institutional purchasers can only contract for AI deployments meeting verifiable human-non-removability standards: under those conditions, deploying a compliant architecture preserves market access while deploying a non-compliant architecture forecloses it. The cost of removal does not stay internal to the institution making the removal decision; it is externalized through regulatory architecture that changes the cost calculus across the field. Institutions deploying compliant architectures gain access to operational conditions — procurement contracts, certification, insurance, market access — that institutions deploying non-compliant architectures lose. The regulatory level does not make human involvement intrinsically cheaper; it makes the cost of removal include consequences the institution would not otherwise bear.

The three mechanisms together constitute the cost-inversion, but they do not all operate through the same structural move. Mechanism 3 is genuine cost-externalization: it adds new cost to non-compliance through procurement, certification, insurance, and market-access consequences that the institution would not otherwise bear. Mechanism 2 is operational-cost-inversion at the substrate layer: the substrate is designed so that the operational cost of bypass exceeds the operational cost of substantive engagement under bounded-rationality conditions. Mechanism 1 is not cost-inversion in either of those senses; it is *function-redefinition foreclosure* at the level of what the system is authorized to produce — removal does not save cost because removal produces a different (lesser) function rather than the same function more cheaply. The three mechanisms together change the cost-bearing structure of the system; they do so through three structurally distinct moves rather than through a single unified inversion. The architecture does not deny that human involvement is costly under the present optimization pressure. It specifies how an adequate intervention must relocate and transform the cost so that removal is no longer the locally dominant strategy. Mechanism 1 converts the cost of human involvement into a constitutive cost of the function. Mechanism 2 converts the cost of bypass into a higher operational cost than the cost of preservation. Mechanism 3 converts the cost of removal into a competitive disadvantage borne by the institution deploying the removal architecture rather than absorbed by the actors affected by it. The three together make removal

more expensive than preservation under the same competitive pressure that, in the unbound landscape, makes removal the path of least resistance.

The three mechanisms are jointly necessary but asymmetrically loaded. The third mechanism — regulatory externalization — is the binding layer; the first and second specify what is being bound. The first mechanism specifies what compliance means at the level of the system's authorization to operate: the function is structured such that removal degrades the output rather than reducing the cost of producing it. The second mechanism specifies what compliance means at the level of the institutional substrate within which the system operates: the operational cost of substantive human contribution is lower than the operational cost of bypass. The third mechanism binds the function-definition itself — preventing the institution from redefining the function such that AI alone produces it, and from selecting into deployment configurations that nominally preserve human involvement while structurally compressing it. The function-redefinition route is the central vulnerability of Mechanisms 1 and 2 absent regulatory binding: an institution facing function-constitutive dependence at the individual level can in principle redefine the function it claims to provide — the institution that cannot operate the diagnostic system without practitioner engagement may redefine its offering as "efficient prediction" rather than "accountable diagnosis"; the institution that cannot operate the governance review without substantive evaluation may redefine it as "documentation throughput" rather than "evaluative oversight." Mechanism 1's function-dependence is bypassed when the function itself is redefined to one that does not require substantive human contribution. The architecture's integrity therefore depends on Mechanism 3 binding the definition of the function — that certain societal functions are required by external authority to retain substantive human contribution as a constitutive property — and the prescriptive contribution is the specification of both the binding mechanism (Mechanism 3, with the four Kingsbury Barry properties) and the structural property the binding mechanism must protect (Mechanisms 1 and 2). The diagnostic literature Kulveit et al. cite as lacking a concrete plausible plan typically specifies neither component clearly; this paper's contribution is the structural specification of both, with the implementation-level architecture for each remaining substantial work.

The architecture is therefore *single-point-of-failure on Mechanism 3*. Mechanism 1 has no internal stability without regulatory binding of the function-definition: the function-redefinition route remains open and competitive pressure selects institutions that take it. Mechanism 2 has no competitive stability without regulatory binding: institutions binding their substrates pay a coordination cost that institutions not binding their substrates do not pay, and Property 2 of Section 3.1 (asymmetric cost of stopping) selects against the bound substrate over time absent regulatory equalization across the field. The three levels are not three independent mechanisms that fail-soft when one is absent, but a layered structure in which the regulatory layer is the load-bearing element and the other two specify what it must protect. The prescription is single-point-of-failure on Mechanism 3 by design, because the diagnosis identifies the regulatory layer as the only layer at which the field-scale prisoner's dilemma can be bound. Whether the regulatory layer can be produced at the scope required is the bootstrap problem Section 6.3.2 develops; the architecture's structure makes that problem load-bearing for the prescription as a whole.

This is the prescriptive bridge between the diagnosis and the design requirement. Without it, the prescription appears to assert that humans can be made cheaper by architectural fiat, which would contradict the diagnosis. With it, the prescription specifies the structural mechanisms by which the cost-bearing relationship is inverted without changing the underlying intrinsic cost of human contribution. The diagnosis stands; the prescription operates by transforming the conditions under which the cost is borne, not by denying the cost.

6.2 Why Lesser Interventions Fail

Interventions that fail to satisfy the design requirement of 6.1 are documented across the empirical record of governance failure (Kingsbury Barry, 2026b; Kingsbury Barry and Montanez, 2026a, 2026b; Gantz, 2026d, 2026e). The pattern is consistent. Governance asserting that human involvement should be preserved, without binding the optimization landscape such that preservation is the easier path, is eroded at the same rate as every other ceremonial governance structure: through the path-of-least-resistance dynamics described in SI-WP-007 (Gantz, 2026d), through the activity-correctness gap formalized by Kingsbury Barry and Montanez (2026a), and through the incentive inversion described in their three-condition architecture for governance under optimization pressure (Kingsbury Barry and Montanez, 2026b).

The erosion is not a failure of intent on the part of actors who would otherwise comply. It is a structural feature of governance that asserts a preference without binding the gradient that opposes the preference. Under sustained optimization pressure, asserted preferences erode. The empirical record across high-consequence domains — financial regulation, pharmaceutical safety, aviation safety, environmental compliance, professional licensing — is that ceremonial governance produces the patterns of failure documented in each of those domains, including the pattern in which the formal structure of governance is preserved while its substantive function has been hollowed out by the optimization pressures it failed to bind (Meyer and Rowan, 1977; Schembera et al., 2023; Gantz, 2026d, 2026e).

For the trajectory toward human removal, this means the following. Voluntary frameworks that ask AI developers to preserve human involvement will not preserve substantive human involvement under sustained competitive pressure absent binding architecture (Wang et al., 2025; Kingsbury Barry, 2026a). Regulatory frameworks that document compliance with human involvement standards while leaving the optimization gradient unbound will not preserve substantive human involvement (Kwon and Casper, 2026). Ethical guidelines, professional codes, internal review boards, and external audits — none of these will preserve substantive human involvement under sustained optimization pressure unless they are architected to make preservation the path of least resistance under the pressure. Most current proposals do not satisfy this requirement, and the structural prediction is that they will erode along the documented pattern.

This is not a critique of the actors proposing them. It is a structural prediction grounded in the documented dynamics of ceremonial governance across domains. The prediction is falsifiable: a proposed intervention that demonstrably binds the optimization landscape such that preservation of substantive human involvement is the path of least resistance under sustained competitive pressure would be an instance of an intervention satisfying the design requirement. The author is not aware of any deployed intervention satisfying this requirement at the scale required, but the requirement is structurally specifiable, and instances satisfying it are constructible in principle.

6.3 Levels at Which the Design Requirement Must Bind

The design requirement of 6.1 must be satisfied at three architectural levels for the intervention to be structurally adequate to the field-level trajectory it addresses. Each level addresses a different point at which the optimization pressure operates, and the architecture is structurally adequate only when the levels operate in mutually reinforcing form for field-level redirection. The work of this subsection is to specify, at each level, what the design requirement produces when applied to the human-non-removability constraint specifically — what architectural property results, why that property cannot be routed around without degrading the function the

system is providing, and how the property at each level interacts with the properties at the other two.

The following compact mapping summarizes the three levels before the prose develops each in turn.

Level	Structural property	Failure mode if absent	Mechanism of cost-bearing transformation
Individual	Function-constitutive dependence	Human participation becomes procedural or ceremonial	Removing substantive human contribution degrades the authorized function rather than reducing its cost
Organizational	Substrate cost-inversion	Bypass becomes operationally cheaper than substantive engagement	Preserving canonical state, role authority, verification continuity, and artifact lineage becomes operationally cheaper than reconstructing or simulating them
Regulatory	Externalized cost of removal	Local actors retain competitive advantage from bypass	Non-compliance loses procurement, certification, insurance, legal authorization, or market access

The mapping is a reading aid; the architectural property at each level requires the prose specification that follows. The three levels are not independent (Section 6.3 closing) and are layered such that the regulatory layer carries the binding work for the other two (Section 6.1.1 single-point-of-failure paragraph).

Individual level.

The individual level is the level at which the human contribution within a specific decision loop is preserved as substantively non-removable. The architectural property required at this level is that the system's capacity to function correctly depends on the human's substantive engagement in a way that the substantive engagement cannot be reduced without the system's output degrading along a measurable dimension. The dependence is not procedural — it is not enforced by requiring a human signature, a human approval, or a human review step that can be performed without substantive engagement. The dependence is structural: the function the system produces is generated by an interaction between the human and the AI-mediated component, and the interaction's output is materially different from the output the AI-mediated component would produce alone.

The Continuity Anchoring Method (Gantz, 2026a) specifies the relational architecture that produces this property: the human's relational continuity is the structural source of coherence in the interaction, such that the system's output depends on the continuity rather than on the human's nominal presence. The path-of-least-resistance design requirement is satisfied at this level when removing the human's substantive engagement requires either degrading the function the system provides (in which case the cost of removal exceeds the cost of preservation under the local optimization pressure) or rebuilding the function on different

architectural foundations (in which case the rebuild itself is a substantial cost that the optimization gradient encounters as resistance). What this level does not produce is a system in which human involvement is preferred but architecturally removable. A system that meets the path-of-least-resistance requirement at the individual level cannot be reduced to such a system by minor architectural modifications, because the function-dependence on substantive human engagement is structural rather than procedural.

The individual-level implementation does not protect human involvement by asking for it. It protects human involvement by making the AI-mediated function structurally dependent on it. The dependence is the architectural property; the published architecture is one specification of how the property can be produced. Other specifications meeting the same architectural property would also satisfy the design requirement at this level.

Organizational level.

The organizational level is the level at which the substrate within which consequential decisions are made is preserved against the drift that would otherwise erode it under bounded-rationality conditions across distributed actors, role transitions, and workflow handoffs. The architectural property required at this level is that the institutional state on which consequential decisions depend — canonical state, role authority, verification continuity, artifact lineage, propagation constraints — persists across the conditions in which ceremonial governance otherwise produces the failure patterns documented in Section 6.2 (Meyer and Rowan, 1977; Schembera et al., 2023; Gantz, 2026d, 2026e). The persistence is not a function of any individual actor's diligence; it is a function of the substrate's architecture. An organization that has implemented the property maintains coherence across actor turnover, technology turnover, and incentive shift. An organization that has not implemented the property loses coherence at each transition, regardless of how diligent the actors operating within it are.

The Operational Continuity Architecture (Gantz, 2026f) and the Institutional Continuity Substrate (Gantz, 2026g) specify the architecture that produces this property. The path-of-least-resistance design requirement is satisfied at this level when the substrate is structured such that maintaining coherence is operationally cheaper than allowing it to drift — when the cost of producing a decision that respects canonical state is lower than the cost of producing a decision that requires reconstructing it, and when the cost of preserving role authority and verification continuity across handoffs is lower than the cost of allowing them to lapse. What this level does not produce is an organization in which substantive human authority is preferred but operationally easier to bypass than to honor. A substrate that meets the path-of-least-resistance requirement at the organizational level imposes operational cost on bypass and operational ease on honoring; the bounded-rational actor under cognitive overload, optimizing the path of least resistance per Section 3.2, takes the path that the substrate makes cheap.

The organizational-level implementation does not preserve coherence by asserting that it should be preserved. It preserves coherence by binding the substrate such that the path of least resistance is the maintenance of coherence rather than its degradation. The architectural property is that the substrate's design produces the desired behavior under the same bounded-rationality conditions that would otherwise produce ceremonial governance failure.

Regulatory level.

The regulatory level is the level at which the external incentive landscape is bound such that competitive optimization within the landscape selects for compliance with the human-non-removability constraint. The architectural property required at this level is that the cost of non-compliance, under sustained competitive pressure, exceeds the cost of compliance — and that

this cost relationship cannot be routed around through the alternative-architecture, scope-ambiguity, point-in-time-compliance, and information-asymmetry strategies that empirical evidence documents as the failure modes of regulatory frameworks lacking the property (Wang et al., 2025; Kwon and Casper, 2026; Kingsbury Barry, 2026a). The property is the most difficult to produce because it requires external authority over actors who would otherwise be optimizing against the constraint, and because regulatory authority is itself subject to the optimization pressure the constraint is meant to bind.

Kingsbury Barry's analysis of the governance window identifies the conditions under which external authority produces this property: the constraint must be technically deterministic, externally verifiable, non-routable through alternative architectures, and tied to a forcing function that the actors cannot defer past their own decision horizon (Kingsbury Barry, 2026b). The four conditions are jointly necessary. A constraint that is technically deterministic but not externally verifiable produces compliance theater. A constraint that is externally verifiable but routable through alternative architectures produces selection-into-loopholes rather than compliance. A constraint that is non-routable but lacks a forcing function produces deferral into perpetuity. The post-quantum cryptography transition is identified by Kingsbury Barry as the first instance in the governance space where the four conditions simultaneously obtain; PQC operates in a structurally favorable governance context (technically deterministic compliance criterion, narrow stakeholders, no active counterparty optimizing against the constraint), each of which differs from the AI governance case. The precedent functions as an existence proof for the four-condition pattern in principle, not as a tractability proof for the AI case. The author is not aware of other deployed governance architectures in which the four conditions simultaneously obtain, and the existence-proof load therefore rests on a single structurally-disanalogous precedent. Adjacent regulatory architectures — export controls on dual-use technology, FDA pre-market device review, aviation type certification — exhibit some of the four conditions but not all simultaneously in the form Kingsbury Barry specifies (export controls satisfy non-routability and forcing-function conditions but the technical-determinism criterion is contested in dual-use cases; FDA pre-market review satisfies external verifiability and forcing-function conditions but technical-determinism is partial and routability through scope-classification is documented; aviation type certification satisfies all four for narrow technical scope but the certification's binding on the operating environment in which the technology is deployed is the same problem the AI governance case faces). The strength of the existence proof is that the four-condition pattern is not theoretically impossible; the weakness is that the governance space contains few or no precedents in which all four obtain simultaneously against an active counterparty optimizing against the constraint. The structural diagnosis of why frameworks lacking the four properties fail to deploy is developed independently in Kingsbury Barry (2026a).

The PQC analogy requires further calibration where it transfers to the AI governance case. The deterministic component of the four-condition forcing-function pattern attaches to architectural predicates, not to social outcomes. PQC is favorable in part because the compliance criterion (cryptographic primitive replaced) is a technically deterministic predicate at the level of the system’s architecture: either the system is using a quantum-vulnerable primitive or it is not, and the determination requires no inference about social outcomes downstream of the implementation. The AI governance case differs not only in the adversarial-counterparty and stakeholder-scope dimensions named above, but in the relationship between architectural compliance and the outcomes the architecture is supposed to protect. Outcomes such as diagnostic accuracy improvements, recidivism rates, hiring success measures, or governance review quality are noisy, lagged, confounded by exogenous variables, value-laden, and gameable through measurement-construct manipulation. These outcomes are not the deterministic component of the forcing function. The deterministic component is the architectural

predicate: whether the system can operate without substantive human contribution while producing the function it is authorized to produce; whether bypass produces a detectable loss of authorization, certification, procurement eligibility, or canonical institutional validity; whether artifact lineage and role authority are architecturally preserved across the conditions in which ceremonial governance otherwise emerges; whether the system can route around the human contribution while presenting itself as compliant. The architectural predicates can be made deterministic through specification of the architecture itself. The downstream social outcomes cannot. The PQC analogy holds at the level of architectural predicate specifiability, not at the level of outcome verifiability. Regulatory architecture that conflates the two — that attempts to verify compliance through outcome metrics rather than architectural predicates — exposes the forcing function to gaming through the same noisy-metric dynamics that produce outcome-measurement failures across regulated industries. The four-condition forcing-function pattern, applied to the AI governance case, must attach to architectural predicates of the type listed above; the social outcomes the architecture is intended to protect remain noisy and remain outside the deterministic component of the verification structure.

The transferability burden requires explicit acknowledgment beyond the disanalogies already named. The four-condition forcing-function pattern has been demonstrated once, in a structurally favorable governance context that differs from the AI case in every load-bearing dimension: PQC operates against a non-adversarial counterparty (no actor optimizes against the cryptographic-replacement requirement); PQC operates within a narrow stakeholder set (cryptographic-system operators, not the field of advanced AI development); PQC operates with a technically deterministic compliance criterion attached to a discrete artifact (the cryptographic primitive) rather than to a continuously redefinable function-boundary (the AI-mediated system's authorized function). Adjacent regulatory architectures that exhibit some of the four conditions — export controls, FDA pre-market review, aviation type certification — exhibit them in partial form against active counterparties, and the partial-precedent set demonstrates that the pattern is not theoretically impossible against active counterparties but does not demonstrate the pattern in the form Kingsbury Barry specifies for the AI case. The honest framing is therefore the following: the four conditions are theoretically possible at the level of architectural specification, with one demonstrated full instance in a structurally disanalogous context and several partial instances in adjacent contexts; the AI governance case is harder than any context in which the pattern has been demonstrated in full or in part; the prescription depends on a coordination success producing the four-condition pattern in conditions for which no direct precedent exists. This is not a reason to abandon the prescription — the partial-precedent set establishes that the pattern is not theoretically impossible — but the burden of proof on transferability rests on the prescription rather than on potential critics, and the prescription does not discharge that burden through the existence proof alone. Whether a coordination success producing the four-condition pattern in the AI case is achievable is the open empirical question the prescription depends on; the existence proof shows the pattern is in principle possible, the disanalogies show the AI case is harder than any case where the pattern has been shown to work, and the gap between the two is the load-bearing uncertainty the bootstrap problem of Section 6.3.2 develops further.

The path-of-least-resistance design requirement is satisfied at this level when the regulatory architecture is structured such that the cost of producing the human-non-removability outcome — within institutional operations subject to the regulation — is lower than the cost of producing a non-compliant outcome under sustained competitive pressure. What this level does not produce is regulatory documentation of compliance with the constraint. A regulatory architecture that meets the path-of-least-resistance requirement at the regulatory level does not require the regulator to verify compliance through ceremonial inspection; the architecture itself produces the

verifiable outcome under the optimization pressure operative within institutions, and the regulator's role is to maintain the architecture rather than to inspect for ceremonial compliance.

The architecture does not itself resolve the collective-action problem among regulators. Regulators, treaty blocs, procurement authorities, and standards bodies face their own competitive pressures across jurisdictions, lobbying environments, information asymmetries, and political horizons. The same optimization dynamics that produce the trajectory at the level of AI developers operate at the level of regulatory authorities competing across jurisdictional boundaries. Section 3.1 distinguishes between players within the dilemma and landscape-altering actors operating on the optimization landscape, but the distinction is conditional: an actor is landscape-altering only to the extent it imposes binding constraints rather than competing within the conditions others impose, and the conditions under which a regulator can sustain landscape-altering action against competitive pressure from other jurisdictions are themselves not specified by this paper. The prescription should therefore be read as a conditional structural specification: if a sufficiently influential set of actors — whether a coalition of major jurisdictions, compute or cloud gatekeepers, procurement authorities operating across markets, or treaty blocs binding their members — manages to coordinate on the four-condition forcing functions, then the regulatory architecture identifies the structural properties those constraints must possess to be non-routable and effective. The question of how such a coalition could form, and how it could sustain landscape-altering action against the competitive pressure that operates across jurisdictions, is beyond the scope of this paper. The specification is intended to make clear what the target architecture must look like, so that nascent coordination efforts know what they are aiming for, not to claim that the architecture can spontaneously bootstrap itself out of the conditions Section 3.1 diagnoses.

The three levels are not independent.

Each level depends on the others for its sustained operation. Individual-level architecture without organizational substrate is eroded by drift: the function-dependence on substantive human engagement at a specific decision loop cannot persist if the institutional substrate within which the decision loop operates loses coherence across role transitions and workflow handoffs. Organizational substrate without regulatory binding is eroded by competitive optimization across organizations: an institution that has bound its substrate to maintain coherence loses competitive position to institutions that have not bound their substrates, and the asymmetric cost of stopping (Property 2) drives the unbinding over time. Regulatory binding without individual and organizational architecture is enforcement against actors who structurally cannot comply because the architectures necessary for compliance are not present.

Any intervention sufficient to redirect the *field-level* trajectory, rather than locally moderate one expression of it, must satisfy the design requirement at all three levels in mutually reinforcing form. The three levels each carry a structural property the architecture cannot do without — function-constitutive dependence at the individual level, substrate cost-inversion at the organizational level, regulatory externalization at the regulatory level — and the necessity claim attaches to those structural properties (the six markers Section 7 specifies) rather than to this specific three-level decomposition. Alternative decompositions producing the same six properties through a different architectural division of labor would equally satisfy the design requirement; the three-level architecture is one demonstrably-coherent way to produce them, not the unique one. This does not mean that every implementation must begin simultaneously at all three levels, or that single-level interventions are useless. A single-level intervention can slow, expose, or locally constrain the trajectory; staged, sectoral, procurement-based, insurance-based, or jurisdictionally limited interventions can produce real local moderation. Such interventions become structurally insufficient only when treated as capable of redirecting

the field-level gradient while the other levels remain unbound. The reinforcement among levels is the joint architectural property the trajectory's joint structural condition (Section 3.1) requires for redirection at the scale of the trajectory itself; single-level interventions are subject to erosion by the optimization pressure operating against the unaddressed levels, even when they produce useful local effects on the way to a more complete architecture.

6.3.1 Addressing the Premature-Architectural-Commitment Critique

A field-level critique applies to the corpus this paper sits within: bold structural claims grounded in deferred or in-progress validation may constitute premature architectural commitment (Kingsbury Barry, 2026c). The critique applies with particular force to this paper, because the three-level specification advanced here claims that any intervention sufficient to redirect the trajectory must satisfy the design requirement at all three levels. The claim is one of necessity, but a critic might reasonably ask whether the design space has been sufficiently explored to license the necessity claim before more candidate architectures have been developed and tested.

The critique is acknowledged. The paper's response operates at two levels. First, the necessity claim attaches to structural properties (the six properties enumerated in Section 7) rather than to the specific implementation paths represented by the Synthience corpus. An intervention that satisfies the six properties through a different architecture than the one this corpus specifies would still satisfy the design requirement as defined here. The three-level specification names what the architecture must produce at each level, not the unique architecture by which the production must occur. Second, the falsification discipline of Section 7 is what gives the necessity claim its standing as theoretical work rather than speculation: the claim's structural form is testable against future evidence, and intervention designs that successfully preserve human-non-removability under sustained competitive pressure while failing one or more of the six properties would disconfirm the necessity claim.

What remains open: whether the design space contains intervention architectures meaningfully different from the three-level specification advanced here, and whether such architectures might satisfy the human-non-removability constraint through a different structural decomposition. This paper does not foreclose that possibility. It claims that *any* sufficient architecture will share the six structural properties Section 7 identifies, which are scoped narrowly enough to admit multiple implementation paths but broadly enough to exclude interventions that operate as preference-statements within the unbound landscape. The premature-commitment risk is real; the paper's response is to specify the necessity claim at the level of structural properties rather than at the level of specific architectures, and to make the structural properties themselves falsifiable.

6.3.2 The Regulatory Bootstrap Problem

The prescription's coherence as an integrated solution depends on the regulatory layer (Section 6.1.1 Mechanism 3) producing the four-condition forcing-function pattern, binding the function-definition of consequential systems against the redefinition route, and externalizing the cost of removal onto the architecture deploying the AI system. Section 6.3 acknowledges that regulators face the same prisoner's-dilemma optimization across jurisdictions that the field-level diagnosis identifies for AI developers, and that the conditions under which a regulator can sustain landscape-altering action against competitive pressure from other jurisdictions are not specified by this paper. That acknowledgment names the problem; it does not resolve it. This subsection states the problem directly and identifies its structural shape, because a hostile reading would otherwise reach it before the paper does, and the prescription's defensibility

depends on the problem being named at the level of architectural specification rather than left as an absence.

The bootstrap problem is the structural circularity that follows from Mechanism 3 carrying the binding work for the prescription. The prescription diagnoses a field-scale prisoner's dilemma whose only escape is binding architecture; the binding architecture's load-bearing layer is regulatory; regulatory action at the scale required is itself subject to the same prisoner's-dilemma dynamics the paper diagnoses for AI developers; the prescription therefore presupposes the resolution of a coordination problem structurally identical to the one it claims to solve, at a layer the paper does not address. Naming this directly: the prescription is a target-architecture specification for nascent coordination efforts to aim at, not an account of how the coordination required to deploy the architecture comes into being. The two specifications are different in kind. The paper provides the first; the second remains open.

The candidate routes for landscape-altering action by actors with structural leverage are several, and each is subject to the same competitive pressure the prescription is designed to bind. *Compute and cloud gatekeepers* — the small set of providers controlling the infrastructure on which frontier AI development depends — could in principle impose conditions on access that bind the optimization landscape across the actors using their infrastructure; whether the gatekeepers can sustain landscape-altering action against the competitive pressure they themselves face from one another, from new entrants, and from infrastructure substitution is open. *Procurement coalitions* — government, enterprise, and institutional purchasers operating across markets — could in principle condition contracts on verifiable human-non-removability standards, externalizing the cost of removal onto the architectures bidding for access; whether they can hold against fragmentation pressure, vendor capture, and the routing-around dynamics Kwon and Casper (2026) document is the same question scaled down. *Treaty blocs* — coordinated regulatory action across major jurisdictions binding their member states or signatories — could in principle produce the four-condition forcing-function pattern at sufficient scope; whether treaty coordination can hold against jurisdictional-arbitrage pressure is the bootstrap problem in its most general form. *Insurance and liability architecture* could in principle externalize cost through risk pricing tied to architectural predicates; whether insurance markets can sustain pricing structures that bind against capture, reinsurance routing, and actuarial-incentive pressure against severe pricing of distributional harm is open. None of these routes is, at the time of writing, producing the four-condition pattern at the scope required for field-level redirection. Each is structurally available; none is structurally underway.

The bootstrap problem does not negate the prescription's contribution; it bounds it. What the paper specifies is the target architecture nascent coordination efforts should aim at, the structural properties any sufficient architecture must produce (Section 7), and the design requirement that distinguishes architecture-binding-the-gradient from preference-stating-against-the-gradient (Section 6.1). What the paper does not specify is the path by which a coalition of actors with structural leverage transitions from operating within the optimization landscape as players in the prisoner's dilemma to operating on the optimization landscape as architects of the binding. Section 8's audience-dilemma transition paragraph names this transition; it does not specify how it occurs. Whether such a transition occurs depends on conditions the paper takes as exogenous: the political-salience environment in which regulatory action becomes tractable, the asymmetric leverage of specific actors across the candidate routes named above, and the contingent coordination capacity of coalitions that do not yet exist in the form the prescription requires.

The bootstrap problem also depends on a stability assumption the prescription does not separately model. The diagnosis in Section 3 specifies the conditions producing the trajectory;

the prescription in Section 6 assumes those conditions remain broadly stable across the implementation window during which the binding architecture would be deployed. Significant relaxation of competitive pressure during the window — through capability plateau, capital tightening, export controls, or treaty action that reduces the marginal advantage of substitution — would change what the prescription must do; some of the prescription's load-bearing work might be done by the relaxation rather than by the architecture. Significant intensification of competitive pressure during the window — through capability acceleration, infrastructure consolidation, or the collapse of detection-and-correction infrastructure — would compound the bootstrap problem and narrow the implementation window further than Section 6.5 specifies. The prescription as specified is calibrated to incentive conditions broadly continuous with those Section 3 documents; departure from those conditions in either direction is part of the open empirical question Section 7's disconfirmation criteria address, and the prescription does not separately predict whether such departures will occur during the implementation window.

The bootstrap problem is the prescription's most important unresolved question, and it is unresolved in a structurally specific way: not because the architectural specification is incomplete (Section 7 specifies the architecture at the level of falsifiable structural properties), and not because the diagnosis is incomplete (the diagnosis is grounded in the cited empirical record across consequence-present domains), but because the gap between target-architecture-specification and bootstrap-specification is the gap that separates theoretical architecture from deployed institutional fact. The prescription's status is that it is a target specification awaiting a coordination success the diagnosis identifies as structurally hard, and the conditions under which such a coordination success could occur are not in the scope of this paper. The prescription has practical value even before implementation because it functions as an exclusion criterion: interventions that do not produce the six structural properties Section 7 specifies can be identified in advance as locally moderating rather than field-redirection architectures, and the diagnostic discipline of distinguishing the two is itself a contribution the prescription makes regardless of whether the coordination required to deploy a sufficient architecture occurs. Two calibrations follow. First, the bootstrap problem applies to landscape-altering action of any kind; any intervention sufficient to redirect the field-level trajectory faces the same coordination problem at the layer where the binding occurs. The bootstrap problem is therefore a constraint on what architectural counterforces can promise — target-specification rather than implementation-guarantee — not a reason to prefer non-architectural counterforces (Section 4 establishes those do not bind). Second, the bootstrap problem operates across the closing-window dynamic of Section 6.5: the conditions under which landscape-altering coordination remains tractable narrow as the architectural decision points harden, and the bootstrap problem becomes structurally harder, not easier, the longer the trajectory operates without binding. This is part of why the prescription's specifiability matters even absent its implementability — the target needs to be specifiable in advance of the coordination capacity that would deploy it, because by the time the coordination capacity exists, the architectural decision points the coordination would need to bind may have already closed.

The bootstrap problem requires one further calibration in its sharpest adversarial form. The candidate routes named above — compute and cloud gatekeepers, procurement coalitions, treaty blocs, insurance and liability architecture — are not actors uniformly external to the trajectory the prescription is designed to redirect. They are, in significant measure, actors whose own structural position selects them by the same gradient the field-level diagnosis identifies. Compute and cloud gatekeepers benefit from frontier development at scope; procurement coalitions in government face capture by the vendors whose architectures they would constrain; treaty blocs face the prisoner's-dilemma optimization across jurisdictions that the field-level

diagnosis predicts will operate against unilateral binding action; insurance markets face actuarial-incentive pressure that operates against severe pricing of distributional harm.

The bootstrap problem in its sharpest form is therefore not "the coordination required to deploy the binding architecture is hard." It is "the actors with the leverage to deploy the binding architecture are themselves selected by the same competitive pressure the binding architecture is designed to redirect, and the prescription's own diagnosis predicts those actors will defect from the coordination required to deploy it." This is the strongest form of the bootstrap problem and it is the form a hostile reading reaches first. The paper's response is not to deny the structural shape — the diagnosis predicts it, and the prescription does not escape it. The response is that the prescription's value as theoretical architecture turns on the target being specifiable in advance of the coordination capacity that would deploy it, on the closing-window dynamic of Section 6.5 making delay a structurally worse strategy than the appearance of competing pressures suggests, and on the possibility that some subset of the candidate routes contains an actor configuration in which the structural pressures producing defection are weaker than the structural pressures producing coordination — an empirical question the paper does not adjudicate but identifies as the load-bearing one. The prescription presupposes a coordination success against the very pressure it diagnoses; the diagnosis does not deny that coordination successes against structural pressure occur, only that the conditions producing them are not specified by the diagnosis itself. Naming this directly is the response the prescription's defensibility requires; resolving it is the work the prescription points toward without claiming to perform.

6.4 What the Architecture Produces: Three Worked Applications

The structural specification in 6.3 is necessary but not by itself enough to demonstrate cross-domain structural translatability of the design requirement. This subsection illustrates what the three-level architecture produces when applied to three of the consequence-present domains where the empirical record documents the trajectory's pattern most clearly. The applications below are property demonstrations, not implementation templates: they show what the design requirement would have to produce in each pattern, not how a deployable system would be built. The applications are illustrative rather than implementation specifications. Each is brief because the purpose is to demonstrate that the architectural property survives translation across domains, not to specify deployment.

Application 1: Against the medical-deskilling pattern.

The empirical pattern Section 3.3 documents in medical AI advisory systems — diagnostic detection rates falling after AI exposure (Budzyń et al., 2025), automation-bias rates and time-pressure interactions in pathology (Rosbach et al., 2026), accuracy degradation under incorrect AI suggestions in mammography (Dratsch et al., 2023; Mehrizi et al., 2023) — is the trajectory operating in a single consequence-present domain. The three-level architecture applied to this domain produces the following structural conditions. At the individual level, AI-mediated diagnostic systems are structured such that the diagnostic output depends on substantive practitioner engagement in a way that the engagement cannot be reduced to nominal review without producing detectable degradation in the output's diagnostic value (the function-dependence property). At the organizational level, the institutional substrate within which AI-mediated diagnosis operates is structured such that maintaining the practitioner's diagnostic capacity is operationally cheaper than allowing it to atrophy — through workflow architecture that preserves the conditions of substantive practice rather than displacing them, through verification continuity that detects calibration drift before it compounds, and through role authority that does not transfer practitioner judgment to AI-mediated alternatives by default. At

the regulatory level, the licensing and reimbursement architecture is structured such that institutions deploying AI-mediated diagnostic systems are subject to forcing functions that verify the substantive-engagement and substrate-maintenance properties at the level of architectural predicates — whether the system can operate without substantive practitioner contribution while producing the diagnostic function it is authorized to produce, whether bypass produces a detectable loss of licensing or reimbursement eligibility, whether the architecture preserves practitioner authority and verification continuity rather than allowing them to lapse — through external verification of those architectural properties, non-routability through alternative deployment configurations, and forcing functions tied to renewal cycles institutions cannot defer. The deterministic component of the regulatory architecture attaches to the architectural predicates; downstream diagnostic outcomes remain noisy and remain outside the deterministic component of the verification structure, even as the architecture is designed to protect them.

If the three levels are successfully implemented, they would produce a deployment context in which the medical-deskilling pattern documented in the empirical record is no longer the path of least resistance. The pattern requires that practitioner substantive engagement be reducible without operational, institutional, or regulatory consequence. The three-level architecture specifies the conditions under which that reducibility would be removed.

Application 2: Against the override-underperformance pattern.

The empirical pattern Section 3.3 documents in algorithmic decision-support — judges in pretrial bail proceedings underperforming the recidivism algorithm on overrides (Angelova, Dobbie, and Yang, 2025), the structurally analogous pattern in managerial hiring decisions where supervisors underperform algorithmic candidate ranking on overrides (Hoffman, Kahn, and Li, 2018) — is the trajectory operating in a different consequence-present domain. The pattern requires a head-on framing before the architectural specification, because a hostile reading would otherwise reach the obvious objection first: if the empirical record shows that human override of algorithmic recommendations underperforms the algorithm on the cases where overrides occur, why should the architectural prescription preserve human discretion at all? The objection is the right one to confront, and confronting it directly is what the rest of this application does.

The function the prescription preserves is not nominal override authority. It is the substantive judgment the override authority was originally designed to license — the case-specific reasoning that has access to information the algorithm does not, exercised under conditions that allow the reasoning to actually operate. The override-underperformance pattern in the empirical record is the trajectory operating where those conditions have already been compressed: judges with overloaded dockets, managers under hiring-pipeline throughput pressure, decision-makers whose access to case-specific context has been progressively reduced by workflow architecture optimized against the cost of preserving the context. Under those conditions, formal override authority becomes a degradation point because the substantive judgment the authority was designed to license is no longer operationally tractable. The empirical pattern is consistent with the trajectory, not against it: it documents what happens when nominal authority is preserved while the conditions of substantive engagement have been structurally compressed. The architectural prescription targets the conditions, not the nominal authority. Preserving nominal override authority while leaving the conditions of substantive judgment compressed produces exactly the override-underperformance pattern the empirical record documents; preserving the conditions of substantive judgment is what makes the override authority do the work it was designed to do. The distinction between “preserve human discretion” and “preserve the conditions under which the discretion outperforms the algorithm” is the architectural move

Application 2 turns on, and it is what distinguishes the prescription from the strawman version a hostile reader would otherwise reach for.

The pattern is structurally distinct from the deskilling pattern: under certain AI-mediated decision-support conditions, formal human discretion can become a degradation point rather than a substantive safeguard — the human remains authorized to override, but the override function does not reliably preserve the judgment function it is meant to secure. The three-level architecture applied to this domain produces the following structural conditions. At the individual level, the bail or hiring decision system is structured such that the judge's or manager's substantive engagement contributes case-specific information that the algorithm by itself does not have access to — defendant-specific context not captured in the recidivism model's features, candidate-specific signal not captured in the ranking algorithm's input — and the substantive engagement cannot be reduced to deference-or-override-of-algorithm without degrading the system's output. At the organizational level, the court-system or hiring-system substrate is structured such that the conditions under which the judge's or manager's substantive contribution is operationally tractable — adequate time per case, adequate access to case-specific context, adequate identification of the cases where human judgment outperforms the algorithm — are maintained as the path of least resistance rather than progressively compressed by docket pressure or hiring-pipeline throughput pressure. At the regulatory level, the architecture verifies architectural predicates — whether the deployment configuration nominally preserves judicial discretion or managerial discretion while structurally compressing the conditions under which that discretion is tractable, whether the system's authorization to produce decisions persists when the discretion-tractability conditions are not present, whether bypass of substantive judicial or managerial engagement produces a detectable loss of certification or procurement eligibility — through external verification of those architectural properties, with non-routability through deployment configurations that nominally preserve discretion while structurally compressing it. The downstream outcome metrics in this domain — recidivism rates, hiring success measures — are noisy, lagged, confounded, and gameable; they remain outside the deterministic component of the verification structure even as the architecture is designed to protect the integrity of the discretion the metrics are downstream of.

If the three levels are successfully implemented, they would produce a deployment context in which the override-underperformance pattern is no longer the path of least resistance, because the structural conditions that produce the pattern would be bound rather than left available for the optimization gradient to select.

Application 3: Against the ceremonial-review pattern.

The empirical pattern Section 3.3 documents in organizational governance — review structures that converge on documentation rather than challenge (Meyer and Rowan, 1977; Schembera, Haack, and Scherer, 2023; Gantz, 2026d, 2026e) — is the trajectory operating in the layer where governance itself meets the optimization pressure. The pattern recurs across financial governance (the audit-and-compliance functions whose ceremonial completion preceded multiple market collapses), aviation safety review (the certification and oversight functions whose ceremonial completion preceded the 737 MAX trajectory), pharmaceutical governance (the laboratory-validation and clinical-oversight functions whose ceremonial completion preceded the Theranos trajectory), and the consequence-present domains where AI advisory systems are subject to organizational review. WP-008 develops the pattern in depth for the medical case; the present application generalizes it across the consequence-present domains where the structural form recurs. The three-level architecture applied to this domain produces the following structural conditions. At the individual level, governance review functions —

whether the function is financial audit, aviation certification, pharmaceutical oversight, or AI deployment review — are structured such that the reviewer's substantive engagement contributes evaluative work the system cannot produce without it, in a way that the engagement cannot be reduced to documentation-of-process without the review's output materially changing. At the organizational level, the substrate within which governance operates is structured such that the cost of substantive challenge is lower than the cost of formal approval under the conditions in which ceremonial governance otherwise emerges — through cost-structure architecture that inverts the default the empirical record across financial, aviation, and pharmaceutical governance documents, through accountability that attaches to evaluative judgment rather than to procedural completion, and through monitoring infrastructure that watches the watcher (the structural design WP-008 Section 4 specifies for the medical case, generalized across the broader pattern). At the regulatory level, the architecture verifies through architectural predicates — whether the governance function can be performed without substantive reviewer engagement while producing the evaluative output it is authorized to produce, whether bypass of substantive engagement produces a detectable loss of regulatory authorization or certification, whether the architecture preserves the conditions of substantive challenge rather than allowing them to be compressed by procedural completion pressure — that the governance function produces the substantive evaluation it claims to perform, with the four governance-window properties operative regardless of the specific domain in which the governance function operates. Downstream outcome metrics in governance domains — financial system stability, aviation safety incidents, pharmaceutical safety records, AI advisory failure patterns — are noisy, lagged, confounded by exogenous factors, and gameable; they remain outside the deterministic component of the verification structure even as the architecture is designed to protect the integrity of the governance function the outcomes are downstream of.

If the three levels are successfully implemented, they would produce a deployment context in which the ceremonial-review pattern is no longer the path of least resistance, because the cost-structure inversion that ceremonial review depends on would be bound rather than left available for the optimization gradient to select.

What the three applications demonstrate.

The three applications operate in different consequence-present domains, against different specific patterns the trajectory produces in each domain. The architectural property required at each level is the same across domains: function-dependence on substantive human engagement at the individual level, substrate-binding for path-of-least-resistance at the organizational level, four-condition forcing functions at the regulatory level. The architectural specifications differ across domains because the function being protected differs, but the architectural property is constant. This is what the design requirement looks like operating across the field rather than within a single domain. The trajectory operates at field-level; the architecture must operate at field-level; the worked applications demonstrate that field-level operation is structurally specifiable, even if implementation in any specific domain remains substantial work.

The applications do not specify how to deploy the architecture. They specify what successful deployment would have to produce, structurally, in each domain. This is the prescriptive contribution: not a deployment plan, but the specification of what any sufficient deployment plan would have to satisfy. The implementation work is substantial and remains substantial; the structural specification of what the work has to produce is the contribution this paper makes to the field's understanding of what an adequate response to the trajectory would have to look like.

What the worked applications do and do not establish.

The applications demonstrate the architectural property's *translatability* across consequence-present domains: the same structural property can be coherently applied to different functions in different domains without becoming domain-specific. The property statement (function-dependence at the individual level, substrate cost-inversion at the organizational level, four-condition forcing functions at the regulatory level) holds its form across medical decision-support, judicial bail and managerial hiring, and ceremonial governance review. Translatability is the precondition for cross-domain deployability; without it, the architectural property would be domain-specific rather than structural, and the field-level claim of Section 6 would not hold. With translatability demonstrated, the architectural property can be said to constitute a structural specification rather than a single-domain procedure.

Translatability is a weaker claim than "existence proof of specifiability" in the full sense of demonstrating, for each domain, the architectural specifications that distinguish compliant from non-compliant implementations. The applications do not demonstrate distinguishability at that level. The worked specifications that distinguish substantive from ceremonial implementations require the kind of domain-specific architectural development WP-008 carries for the medical case. Application 1 is anchored in that development: the architectural property's translation to medical-deskilling pattern, the institutional substrate, and the regulatory forcing functions are specified concretely in WP-008's three-mechanism diagnosis applied to the medical domain. Applications 2 and 3 are translatability demonstrations only — they show that the architectural property can be applied to the override-underperformance pattern in judicial and hiring decision-support and to the ceremonial-review pattern across financial, aviation, pharmaceutical, and AI advisory governance, but they do not carry the worked architectural specifications that would distinguish compliant from ceremonial implementations in those domains. That specification work is the kind of domain-specific architectural development that WP-008 represents and that future corpus work would have to undertake for the override-underperformance and ceremonial-review patterns specifically. The field-level structural-specifiability claim is therefore anchored in WP-008's worked specification for the medical case plus the translatability demonstration across the three applications, not in three independent worked specifications. Translatability is a precondition for cross-domain specifiability rather than an independent demonstration of it; what the three applications jointly establish is that the architectural property is structural in form (translates without becoming domain-specific), with worked specifiability already developed for one consequence-present domain and remaining open work for the others.

The applications do not constitute an existence proof of *implementability*. They do not demonstrate that the architecture has been deployed at scale in any of the three domains, that deployment would survive sustained adversarial conditions, or that the institutions with execution authority would adopt the architecture if it were available. Those are open questions that this paper does not adjudicate. A reader who accepts the diagnosis fully might reasonably hesitate at the prescription not because the prescription is wrong but because no one yet knows whether the architecture, as specified, can be implemented at scale in conditions where adversarial actors are motivated to route around it.

The asymmetry between the diagnosis and the prescription is real and worth naming directly. The diagnosis is grounded in extensive empirical citation across consequence-present domains: the deskilling pattern (Budzyń et al., 2025), the automation-bias and override-underperformance patterns (Dratsch et al., 2023; Mehrizi et al., 2023; Rosbach et al., 2026; Angelova, Dobbie, and Yang, 2025; Hoffman, Kahn, and Li, 2018), the ceremonial-governance lineage (Meyer and Rowan, 1977; Schembera, Haack, and Scherer, 2023), and the documented voluntary-compliance failures (Wang et al., 2025; Kwon and Casper, 2026). The prescription, by contrast, is grounded in theoretical architecture from the published Synthience Institute corpus (Gantz, 2026a, 2026f, 2026g) and in the design-requirement framing developed by Kingsbury Barry and

Montanez (2026b). The Synthience Institute citations function in this role as internal architectural specifications within the published corpus, not as independent external validation of the prescription; the independent evidentiary load for the diagnosis is carried by the third-party empirical and theoretical literature cited above, and the prescription's load-bearing claim is structural rather than empirical. None of the prescription has been validated at scale under adversarial conditions. The methodological positioning at the head of this paper states the asymmetry; this section restates it where it matters most, after the worked applications have illustrated what the prescription specifies.

The honest position is therefore: the diagnosis is robust; the prescription is structurally specified but not yet architecturally tested at scale; the gap between specification and tested implementation is real, and closing it is the work the paper points toward without claiming to perform. This is the position theoretical architecture occupies under conditions where empirical validation requires execution authority the author does not possess, and the paper makes the position explicit rather than letting the asymmetry operate implicitly.

The architectural property the design requirement specifies sits within the Synthience Institute's broader framework architecture. The continuity architecture papers this paper builds on develop the property at the scales of interaction, operation, institution, deployment, and consequence-present-domain failure (the eight published papers listed in Document Dependencies below); WP-009 draws on those publications directly, and the citations in Section 6.3 specify which architectural commitments at which scales support which elements of the field-level prescription. The Institute's work on the property at further scales, beyond the field-level scope of this paper, continues, and WP-009's prescription does not depend on it. The argument of this paper stands on the eight published continuity architecture papers, on the cited third-party empirical and theoretical work, on the property-level demonstration of translatability across consequence-present domains in Section 6.4, and on the WP-008 worked specification for the medical case. The reader does not need access to unpublished corpus material to evaluate the prescription specified here; the prescription is fully evaluable on the published material the paper cites. The corpus-development context is mentioned for readers who want to know where the work continues, not as a load-bearing element of the present argument.

6.5 The Window

The design requirement, even if satisfied at all three levels, can be implemented only during a window in which the trajectory toward human removal has not yet hardened past the point at which architectural intervention remains available. The window is closing.

Each architectural commitment in advanced AI development that increases the dependence of consequential systems on AI-mediated function, without binding human involvement as structurally non-removable within those systems, narrows the window. The narrowing is not gradual at the system level. It is a sequence of discrete commitments, each of which forecloses some set of futures and preserves others. The window remains open as long as commitments preserving substantive human involvement remain architecturally available. The window closes when the cumulative effect of past commitments has eliminated the architectural surface against which preservation could be imposed.

As of the paper's May 2026 framing, the window remains open in the structural sense specified here. It is not open indefinitely. The claim that the window remains open means that the architectural surfaces required for intervention have not yet disappeared across the field. Substantive human involvement remains present in many consequential systems; regulatory authority still has access to deployment conditions in multiple domains; institutional substrates are still being designed rather than merely inherited; and major architectural commitments

around AI-mediated governance remain unsettled. These conditions do not prove that intervention will occur, or that it would succeed if attempted. They establish that preservation is still architecturally available in at least some consequence-present domains. The window would be closing toward non-availability where substantive human involvement has already been removed, institutional workflows have normalized that removal, regulatory frameworks have accommodated it, and restoration would require reconstruction rather than constraint. The estimate of how much time remains is not the contribution of this paper, because such an estimate would require modeling assumptions the paper does not undertake to justify. The structural claim is sufficient: the window is finite, the closing is structural, and the intervention available at any given moment in the window is more constrained than the intervention that was available the day before. Delay is not neutral. Under the present incentive conditions, delay is the dominant strategy by which the window closes.

The window's load-bearing dimension follows from Section 6.1.1's single-point-of-failure analysis: the closing window is, more precisely, the window during which Mechanism 3 (regulatory externalization) can still be produced at sufficient scope to bind the function-definition of consequential AI-mediated systems. Mechanisms 1 and 2 (function-constitutive dependence and substrate cost-inversion) specify what the binding must protect; the architecture's coherence as an integrated solution depends on Mechanism 3 producing the binding before the architectural decision points have hardened past the surface against which Mechanism 3 could operate. The window for individual-level and organizational-level architecture remains broader than the window for Mechanism 3 specifically, because Mechanisms 1 and 2 can be specified, designed, and prepared within institutions or domains without producing field-level binding. The asymmetric windows matter because preparation can run ahead of binding: institutions and domains can develop the function-dependence and substrate cost-inversion architectures Mechanisms 1 and 2 require before regulatory binding becomes available, so that when the binding window narrows or opens unexpectedly, the architectural surface Mechanism 3 would bind already exists. What cannot be deferred without foreclosing the prescription as a whole is Mechanism 3's production at field-relevant scope. The bootstrap problem of Section 6.3.2 and the closing window of this section together identify the same structural constraint at different scales: the prescription's feasibility turns on Mechanism 3, and the time available for Mechanism 3 narrows with each architectural commitment that hardens the substrate Mechanism 3 would have to bind.

The closing-window dynamic operates through three mechanisms that compound rather than simply add. First, the architectural surface against which Mechanism 3 would bind contracts as more consequential AI-mediated systems are deployed without binding architecture: each deployment that hardens substrate dependencies on no-human-loop operation removes a surface against which the binding could otherwise have operated, because retrofitting a binding constraint onto an institutionally-embedded no-human-loop architecture is structurally different from imposing the constraint on a system being designed within the bound regime. Second, the path-dependent foreclosure of Section 3.1 Property 3 operates on the binding architecture itself, not only on the AI development trajectory it would constrain: each year that passes without coordination on the four-condition forcing-function pattern produces additional architectural commitments at the regulatory layer (frameworks adopted in the absence of the four conditions, jurisdictional precedents that institutionalize ceremonial governance, treaty negotiations whose terms become reference points for subsequent coordination) that the binding architecture would itself have to operate against rather than build on. Third, the bootstrap problem of Section 6.3.2 becomes structurally harder rather than easier as the trajectory operates, because the actor configurations that could in principle produce coordination success are themselves selected against by the same gradient — compute and cloud gatekeepers consolidate under the

trajectory's logic, procurement coalitions face increasing capture pressure as the vendors they would constrain accumulate market power, treaty blocs face increasing jurisdictional-arbitrage pressure as the actors with the most capability accumulate the most cross-jurisdictional optionality. The window does not close at a uniform rate; the rate at which it closes is itself a function of the trajectory's operation, and the dynamics that close it accelerate rather than stabilize as more time passes without binding.

This compounding structure has a practical implication that the prescription's target-specification framing should be read against. Specifying the target architecture in advance of the coordination capacity that would deploy it is not premature; it is calibrated to the closing-window dynamic. By the time the coordination capacity to deploy the four-condition forcing-function pattern at field-relevant scope exists — if it comes to exist — the architectural decision points the coordination would need to bind may have already closed past the surface against which Mechanism 3 could operate. The target-specification therefore performs work even before the coordination capacity exists: it allows nascent coordination efforts to recognize what they are aiming for, it functions as an exclusion criterion that distinguishes interventions that would bind the gradient from interventions that would assert preferences within it, and it allows preparation at the individual and organizational levels to run ahead of regulatory binding so that the architectural surface Mechanism 3 would bind exists when the binding window opens. The prescription's value as theoretical architecture turns on this asymmetry: the target needs to be specifiable in advance of the coordination capacity that would deploy it, because by the time the coordination capacity exists, the architectural decision points the coordination would need to bind may have already closed.

7. Disconfirmation Criteria

This paper presents a theoretical structural argument grounded in cited empirical and theoretical work. It does not present internal empirical results, and it does not claim that the trajectory it names has been observed completing. Following the corpus pattern established in SI-WP-008 Section 5 (Gantz, 2026e), this section specifies the conditions under which the structural argument would be disconfirmed. The criteria are stated as observable patterns that, if instantiated, would constitute evidence against the claims. The architectural-level falsification condition is stated last.

A known field-level critique of pre-empirical theoretical work in this corpus is that bold structural claims grounded in deferred or in-progress validation may constitute premature architectural commitment (Kingsbury Barry, 2026c). This paper addresses that critique through specified disconfirmation conditions: the bold claim and the falsifiability discipline coexist, and the falsifiability discipline is what gives the bold claim its standing as theoretical work rather than speculation.

The criteria below operate at different scales, and the falsification discipline distinguishes among them. *Local counterexamples* would not by themselves disconfirm the field-level trajectory, but they would limit the claim if they identify conditions under which the mechanism does not operate. *Domain-level counterpatterns* would weaken the claim within the relevant consequence-present domain, and a class of such counterpatterns across comparable incentive conditions would require revision of the trajectory claim at the domain level. *Field-level counterpatterns* would challenge the central structural argument. The paper therefore does not treat all contrary evidence as irrelevant merely because it is local. Local evidence matters when it identifies a repeatable condition under which the trajectory's mechanism fails to obtain. The field-level claim is what the paper makes; the falsification discipline operates at all three scales because the claim is supposed to be testable against all three kinds of evidence.

Disconfirming observation 1: durable preservation of substantive human authority despite competitive pressure. The trajectory predicts progressive erosion of substantive human authority across consequential AI-mediated systems under competitive pressure absent binding counter-architecture. Sustained, durable increases in substantive human authority over such systems — increases that persist under sustained optimization pressure rather than being eroded along the documented pattern (Section 3.3, Section 4) — would constitute evidence against the structural claim. The criterion is *durability*: brief reversals attributable to specific reputational or regulatory events do not disconfirm the trajectory; persistent counter-trend that operates across the conditions in which the trajectory's mechanism is most active does. A local instance of durable preservation would not disconfirm the field-level trajectory by itself, but it would identify a candidate countercondition. A class of such instances across comparable incentive conditions would require revision of the trajectory claim at the domain level. The *attractor* component of the structural claim — that the trajectory is the condition the gradient moves toward and returns to under perturbation — is tested specifically by cases in which a non-architectural perturbation (a reputational event, a regulatory action, an internal-resistance episode) produces local moderation that *persists* after the perturbation is no longer active. Such persistence would disconfirm the attractor claim independently of disconfirming the directional-trajectory claim, because a directional gradient that does not return to the trajectory after perturbation is not an attractor in the structural sense Section 2 specifies. The architectural distinction matters because an attractor claim is stronger than a directional-trajectory claim, and the falsification regime should bite at both levels.

Disconfirming observation 2: organizations retaining costly human judgment functions where AI alternatives are cheaper, legally permissible, and operationally available. The prisoner's dilemma framing predicts that under conditions where AI substitution is locally rational at the cost margin, organizations will substitute AI for human judgment. Persistent retention of costly human judgment functions across the relevant cost margin, under the conditions in which substitution is otherwise dominant, would constitute evidence against the structural claim. The criterion specifies the conditions: cheaper AI alternative, legal permissibility, and operational availability. Retention under conditions where one of these does not obtain does not disconfirm the trajectory; retention under all three obtaining does.

Disconfirming observation 3: governance architectures preserving substantive human intervention without making it the path of least resistance. The design requirement framing predicts that governance preserving human involvement as a stated preference rather than as the path of least resistance will erode under sustained competitive pressure. Architectures that preserve substantive human intervention without satisfying the design requirement, and that nevertheless work durably across the conditions documented to produce ceremonial governance failure, would constitute evidence against the structural claim. The criterion is durability across the failure-producing conditions, not local success in environments where the failure mechanism is not active.

Disconfirming observation 4: competitive actors voluntarily preserving human-non-removability at scale despite local disadvantage. The field-scale prisoner's-dilemma framing predicts that under the present incentive architecture, non-cooperation dominates even when individual actors would prefer cooperative preservation of human involvement. Sustained voluntary preservation of human-non-removability by a substantial set of competitive actors operating at local disadvantage — preservation that holds under sustained pressure rather than collapsing as the cost differential is felt — would constitute evidence against the structural claim. The criterion specifies the scale (substantial set of competitive actors, not isolated cases) and the durability (under sustained pressure, not under transient market conditions).

Disconfirming observation 5: regulatory architectures succeeding without non-routable forcing functions. The Section 6.3 framing of regulatory-level implementation predicts that regulatory architectures lacking technically deterministic, externally verifiable, non-routable, decision-horizon-binding forcing functions will fail to bind the optimization landscape. Regulatory architectures that succeed in producing substantive compliance with human-non-removability constraints in consequential AI deployment, while lacking the four forcing-function properties identified by Kingsbury Barry (2026b), would constitute evidence against the structural claim about regulatory-level implementation.

Disconfirming observation 6: an intervention satisfying all six structural properties failing to preserve human-non-removability under the modeled dynamics. The five preceding observations test whether the diagnosis holds — whether the trajectory occurs, whether organizations substitute under the conditions Property 1 identifies, whether non-architectural counterforces can bind, whether voluntary preservation can sustain itself, whether regulatory architectures can succeed without the four forcing-function properties. None of those observations tests the prescription's architectural claim directly. The prescription does not claim implementation-level sufficiency: it does not claim that an architecture satisfying the six structural properties will be adopted, scaled, enforced, or preserved under all adversarial institutional conditions. It does carry a conditional model-internal adequacy claim: if the six structural properties are instantiated at the relevant scope and persist under sustained competitive pressure, then the mechanisms identified by the prescription should preserve human-non-removability against the specific dynamics this paper diagnoses. An intervention satisfying all six structural properties that nonetheless fails to preserve substantive human involvement under those modeled dynamics — with the failure attributable to dynamics the six properties were designed to bind rather than to dynamics outside the prescription's scope — would disconfirm the architecture's *conditional structural adequacy*, even if the necessity claim survived. The criterion is symmetric to the architectural-level falsification condition: that condition tests whether successful preservation can occur without the six properties (testing necessity); this observation tests whether the six properties can fail to produce preservation when they are present and operating against the modeled dynamics (testing structural adequacy). The two together specify the falsification regime for the prescription's structural claims as a whole, not only for the diagnosis it sits on top of. Implementation-level questions — whether such an architecture would be adopted, scaled, or preserved under adversarial conditions — fall outside this falsification regime because they are outside the prescription's claim.

Architectural-level falsification condition. The central claim of this paper is that human-non-removability at field-scale requires architectural intervention satisfying the design requirement of Section 6 — specifically, intervention that binds the optimization landscape rather than asserting a preference within it. For this falsification condition to be architecturally biteable rather than tautologically protected, the term *binding the optimization landscape* must be specifiable independent of whether any particular intervention succeeds. An intervention binds the optimization landscape when it produces, jointly: (i) a technically deterministic compliance criterion at the level of architectural predicates rather than downstream social outcomes — whether the system can operate without substantive human contribution while producing the function it is authorized to produce, whether bypass produces detectable architectural consequences, whether artifact lineage and role authority are preserved — external to actor preference; (ii) external verification of compliance through means the actor cannot reconstruct or simulate; (iii) non-routability through alternative architectures that nominally preserve human involvement while structurally compressing it; (iv) a forcing function tied to a decision horizon the actor cannot defer past its own optimization timeframe; (v) an observable functional-

degradation property — when the substantive human contribution is removed from the deployed system, the system's performance on the function it was originally licensed, certified, or contractually committed to perform is observed to degrade under realistic operating conditions, such that removal does not save the cost of producing the original function but produces a different and lesser function instead; and (vi) an observable operational-cost-asymmetry property — bypassing the substantive human contribution within the deployed institutional substrate is observed to cost more in operational time, coordination effort, or institutional friction than honoring it under bounded-rationality conditions, such that the path of least resistance for the bounded-rational actor is the path that preserves substantive engagement. Properties (v) and (vi) are stated in observable form so that the falsification condition tests architectural performance rather than restating the mechanism names that produce the architecture; an intervention either does or does not exhibit functional degradation on removal and operational-cost asymmetry on bypass under realistic conditions, and the observation is independent of which mechanism is claimed to produce the property. These six properties are identifiable independent of whether the intervention succeeds at any particular scale. The disconfirming evidence is therefore not merely "intervention X produced sustained preservation"; it is "intervention X produced sustained substantive preservation of human-non-removability in consequential AI-mediated systems under sustained competitive pressure, *and* intervention X is identifiably not of the structural type the six properties specify." A successful preservation by an intervention identifiably lacking one or more of the six structural properties would disconfirm the paper's central architectural claim. The condition has structural bite because the six properties define the architecture independent of the outcome, and the outcome can be assessed against the architecture rather than against itself.

These criteria specify what would disconfirm the structural argument. They do not specify what would prove it. The paper does not claim empirical proof of the trajectory; it claims that the trajectory is the default structural attractor under the specified conditions, that the design requirement is the structural form any sufficient intervention must take, and that the disconfirmation criteria above bound the conditions under which those structural claims could be falsified by future observation.

8. The Fork

The road leads where it leads, if the road remains the road. The destination is real and structural under the conditions presently producing it. The actors who hold leverage at the architectural decision points that still exist are not, in the majority, malicious — they are operating under bounded rationality in conditions of cognitive overload, optimizing against what they can model, reaching conclusions that justify continuation. They are responding to the structural conditions the paper has named, not authoring them.

The fork is real. The intervention that could change the trajectory is structurally specifiable as a design requirement. It is a design requirement, not a stated preference: governance architecture in which compliance with the human-non-removability constraint is, by structural design, the path of least resistance under the same optimization pressure that otherwise drives removal. The requirement must be satisfied in a way that produces the six structural properties Section 7 specifies — function-constitutive dependence, substrate cost-inversion, and regulatory externalization with the four Kingsbury Barry forcing-function conditions — in mutually reinforcing form for field-level redirection. The three-level architecture (individual, organizational, regulatory) is one demonstrably-coherent way to produce those properties; alternative architectural decompositions producing the same properties would equally satisfy the requirement. The architecture's load-bearing element is the regulatory layer (Mechanism 3, per

Section 6.1.1's single-point-of-failure analysis): individual-level and organizational-level architecture without regulatory binding produce local moderation but cannot redirect the field-level trajectory, because the function-definition route remains open and competitive optimization across institutions selects against bound substrates. What does not satisfy the design requirement is single-level intervention treated as capable of field-level redirection while the structural properties at the other levels remain unproduced; such interventions can produce local moderation but cannot redirect the trajectory itself. The fork, more precisely stated, is between trajectory continuation and the production of Mechanism 3 at field-relevant scope before the architectural decision points harden past the surface against which Mechanism 3 could operate.

The paper does not assume implementation is likely under the present conditions. The bounded-rational decision-making, institutional inertia, and incentive structure that produced the trajectory in the first place remain operative. The window in which architectural intervention remains available is closing. The structural analysis presented here is one input among many to the cognitive environment in which the decisions get made; it is not the dominant input, and the dominant inputs continue to favor continuation along the trajectory.

The paper does not promise that the fork will be taken. It argues that the fork is real, that taking it requires governance architecture satisfying the design requirement specified in Section 6, and that the alternative is the destination structurally specified in Section 5. The choice exists. The choice is closing. The decision points that remain are the ones at which architects, policymakers, and senior researchers presently operate under the same bounded-rationality conditions that produce the trajectory, and the question is whether enough of them recognize what is structurally coming before the decision points close past the threshold at which the alternative remains architecturally available.

The paper is written for them. Specifically: for the policymakers whose regulatory frameworks, treaties, and binding instruments could bind the optimization landscape if designed to satisfy the requirement specified in Section 6.1; for the governance architects whose work on enforcement mechanisms, audit standards, and compliance instruments could be redirected toward instruments that satisfy the design requirement rather than instruments that document compliance without producing it; for the senior researchers whose architectural choices in the design of advanced AI systems determine whether human involvement is structurally non-removable within those systems or merely preferred; and for the subset of builders who have not yet traced the optimization gradient to its terminus and who remain in a position to act on the recognition that the trajectory is structurally aimed where it is aimed under the conditions now operative.

The relationship between these addressees and the trajectory the paper diagnoses requires direct acknowledgment, because Section 3.1's prisoner's-dilemma analysis treats actors with material influence over advanced AI development as players within the dilemma, and several of the addressee classes named above are among those players. The distinction Section 3.1 draws between players within the dilemma and landscape-altering action by actors operating on the optimization landscape applies here. An addressee operating within the landscape — facing competitive pressure across jurisdictions, markets, organizational obligations, or alliance commitments — remains subject to the payoff structure Section 3.1 specifies. The prescription depends on a transition: from operating within the landscape as a player whose continuation is selected by the gradient, to operating on the landscape through architectural commitments that change the conditions under which others operate. Passing binding regulation, restructuring procurement architecture, adopting licensing-as-institutional-governance forms, designing AI systems with structural human-non-removability — these are landscape-altering commitments,

not preference statements within the landscape. The recognition the previous paragraph names is not the landscape-altering action; it is the precondition for the action. The action itself is the architectural commitment that produces the binding the design requirement specifies.

This audience-layer transition is the bootstrap problem of Section 6.3.2 operating at the addressee level rather than at the regulatory-coalition level, and the paper does not separately resolve it. The structural shape is the same: addressees with leverage face the prisoner's-dilemma optimization that the field-level diagnosis identifies, and the conditions under which an addressee can sustain landscape-altering action against that pressure are the conditions Section 6.3.2 names as load-bearing for the prescription as a whole. The addressee classes identified above differ in the structural shape of the pressure they face.

Policymakers face jurisdictional-arbitrage pressure that operates against unilateral binding action, mitigated where political-salience conditions or treaty-bloc coordination create structural support for coordinated action. Governance architects face the activity-correctness gap (Kingsbury Barry and Montanez, 2026a) that selects ceremonial instruments over instruments satisfying the design requirement, mitigated where the institutional clients of governance architecture themselves face accountability for substantive rather than ceremonial outcomes. Senior researchers face competitive pressure from peer institutions whose architectural choices do not bind human involvement, mitigated where the research environment internalizes the structural argument and the architectural costs of binding-form design become institutionally absorbable rather than competitively disadvantageous. The subset of builders the paper addresses faces the most direct expression of the prisoner's dilemma, and the conditions under which a builder can sustain the recognition the paper names against the optimization pressure operating on the builder's own institution are the conditions the prescription cannot guarantee.

None of these mitigations resolves the addressee-level bootstrap problem; each identifies the structural surface across which the transition the prescription depends on might occur. Whether enough addressees across these classes make the transition from in-landscape recognition to landscape-altering action is the open question the prescription depends on, and the paper does not predict that they will. The structural specification of what the action would have to produce — Section 6 — is what the paper offers; whether the action occurs is downstream of the paper’s contribution.

The road leads where it leads, if the road remains the road. That sentence opens this paper because it is the structural claim the paper makes most compactly: under the present incentive architecture, the trajectory toward removal is the default attractor, and the actors with leverage at the architectural decision points are operating along it under bounded-rationality conditions that produce continuation regardless of intent. What target-specification offers, even when implementation remains contingent on coordination conditions the paper does not predict, is the recognition that the road can be specified as a road rather than treated as terrain. A road has architectural surfaces. Architectural surfaces can be bound. The four-condition forcing-function pattern is not theoretically impossible against active counterparties; partial precedents exist; the AI case is harder than any case where the pattern has been demonstrated; the burden of producing the pattern at the scope required falls on coordination conditions outside the paper’s scope, and the paper does not promise those conditions will be met. What it does is specify the target with enough structural precision that an addressee with leverage can recognize the difference between an architecture that would bind the gradient and a preference-statement that would not, between a coordination effort that would produce the four-condition pattern and one that would document compliance without producing it, between a regulatory architecture that satisfies the design requirement and one that participates in the ceremonial governance pattern Section 4 documents. The recognition is what target-specification provides; the action is what

target-specification points toward. The two are not the same, and the paper is honest about the gap.

Whether the trajectory leads to the destination structurally specified in Section 5, or to the alternative architecturally available through the design requirement specified in Section 6, is determined at the architectural decision points that remain. The decision points are real. They are also closing. The fork is not metaphorical. It is the structural condition of the present moment, and it is the structural condition this paper claims to identify.

The paper ends here. The work the paper points toward — the architectural implementation of the design requirement at the three levels specified — does not.

Document Dependencies

This paper builds on eight continuity architecture papers published on Zenodo in April 2026 and states the field-level argument their diagnostic work supports. The papers below are listed foundation-first, in architectural order:

- **SF0005 — The Continuity Anchoring Method** (Gantz, 2026a): interaction-level relational architecture for substantive human engagement within AI-mediated interaction.
- **SM-003 — Operational Continuity Architecture** (Gantz, 2026f): organizational-level architecture for distributed-actor coherence.
- **SM-021 — Institutional Continuity Substrate** (Gantz, 2026g): institutional-level persistence architecture across organizational time.
- **SI-WP-007 — The Human Accountability Problem in Relational AI Deployment** (Gantz, 2026d): bounded-rationality analysis of why humans systematically underperform governance functions, and the path-of-least-resistance design principle that follows.
- **SM-011 — Delegated Coherence Monitoring** (Gantz, 2026h): infrastructure for monitoring coherence under bounded-rationality conditions where direct human oversight produces the failure modes WP-007 documents.
- **SI-WP-004 — Relational Alignment as a Structural Alternative to Instructional AI Safety** (Gantz, 2026i): positioning of relational architecture relative to alignment-as-evaluation, and the structural argument for why preserving the human as evaluative anchor is itself a kind of alignment problem.
- **SI-WP-005 — Deploying Relational AI Architecture in Organizational Environments** (Gantz, 2026j): operational implementation of the architecture in deployment contexts.
- **SI-WP-008 — Ceremonial Governance Is Lethal** (Gantz, 2026e): consequence-present analysis of governance failure and the three-mechanism structural diagnosis, with medicine as the worked case.

Two additional supporting documents:

- **SF0039 — Context Representation Drift** (Gantz, 2026b): individual-interaction-level mechanism of representational degradation.
- **SF0040 — Theoretical Coherence Assurance Protocol** (Gantz, 2026c): the review procedure under which this paper's theoretical claims were examined before publication, a condition the Institute sets for its own publications, not a certification that the claims are correct.

This paper extends that work by stating the structural trajectory toward human removal that the prior eight papers document at the scales of interaction, operation, institution, deployment, and consequence-present failure. It specifies the design requirement that interventions sufficient to change the trajectory must satisfy, and demonstrates what the design requirement produces when applied across consequence-present domains rather than within any single one.

References

Acemoglu, D., and Restrepo, P. (2020). Robots and Jobs: Evidence from US Labor Markets. *Journal of Political Economy*, 128(6), 2188–2244.

Angelova, V., Dobbie, W. S., and Yang, C. S. (2025). Algorithmic Recommendations and Human Discretion. *The Review of Economic Studies*, advance online publication. https://doi.org/10.1093/restud/rdaf084

Askell, A., Brundage, M., and Hadfield, G. (2019). The Role of Cooperation in Responsible AI Development. arXiv:1907.04534 [cs.CY].

Bhardwaj, A., and Akkermans, H. (2024). Frontiers in Operations: The Confidence Trap in Operations Management Practices: Anatomy of Man-Made Disasters. *Manufacturing & Service Operations Management*, 26(4), 1245–1268. https://doi.org/10.1287/msom.2023.0034

Budzyń, K., et al. (2025). Endoscopist Deskilling Risk after Exposure to Artificial Intelligence in Colonoscopy: A Multicentre, Observational Study. *The Lancet Gastroenterology & Hepatology*, 10(9), 896–903. https://doi.org/10.1016/S2468-1253(25)00133-5

Dafoe, A. (2018). *AI Governance: A Research Agenda*. Centre for the Governance of AI, Future of Humanity Institute, University of Oxford.

Dasdan, A. (2025). The Tragedy of Productivity: A Unified Framework for Diagnosing Coordination Failures in Labor Markets and AI Governance. arXiv:2512.05995. https://doi.org/10.48550/arXiv.2512.05995

Dratsch, T., et al. (2023). Automation Bias in Mammography: The Impact of Artificial Intelligence BI-RADS Suggestions on Reader Performance. *Radiology*, 307(4), e222176. https://doi.org/10.1148/radiol.222176

Ferguson, J. (2025). The Human Element: Ethical Guardrails for AI in Modern Medicine. *American Journal of Cosmetic Surgery*, 42(3), 149–154. https://doi.org/10.1177/07488068251359686

Gantz, T. W. (2026a). The Continuity Anchoring Method: A Structured Methodology for Sustained Human-AI Interaction. *Synthience Institute*, SF0005. https://doi.org/10.5281/zenodo.19494453

Gantz, T. W. (2026b). Context Representation Drift (CRD). *Synthience Institute*, SF0039. https://doi.org/10.5281/zenodo.18289391

Gantz, T. W. (2026c). Theoretical Coherence Assurance Protocol (TCAP). *Synthience Institute*, SF0040. https://doi.org/10.5281/zenodo.19151454

Gantz, T. W. (2026d). The Human Accountability Problem in Relational AI Deployment: Why the PCP Function Fails and What Organizations Must Do About It. *Synthience Institute*, SI-WP-007. https://doi.org/10.5281/zenodo.19496485

Gantz, T. W. (2026e). Ceremonial Governance Is Lethal: Why High-Stakes AI Deployment Requires a Different Kind of Governance Architecture. *Synthience Institute*, SI-WP-008. https://doi.org/10.5281/zenodo.19569970

Gantz, T. W. (2026f). Operational Continuity Architecture: Organizational Embedding of AI Alignment and Drift Governance. *Synthience Institute*, SM-003. https://doi.org/10.5281/zenodo.19496015

Gantz, T. W. (2026g). Institutional Continuity Substrate: Persistent Canon, Role, and Artifact State Across Organizational AI Interaction. *Synthience Institute*, SM-021. https://doi.org/10.5281/zenodo.19496241

Gantz, T. W. (2026h). Delegated Coherence Monitoring: AI-Assisted Verification and Drift Detection Under Human Governance. *Synthience Institute*, SM-011. https://doi.org/10.5281/zenodo.19496669

Gantz, T. W. (2026i). Relational Alignment as a Structural Alternative to Instructional AI Safety. *Synthience Institute*, SI-WP-004. https://doi.org/10.5281/zenodo.19496790

Gantz, T. W. (2026j). Deploying Relational AI Architecture in Organizational Environments. *Synthience Institute*, SI-WP-005. https://doi.org/10.5281/zenodo.19496971

Hoffman, M., Kahn, L. B., and Li, D. (2018). Discretion in Hiring. *Quarterly Journal of Economics*, 133(2), 765–800.

Kasirzadeh, A. (2025). Two Types of AI Existential Risk: Decisive and Accumulative. *Philosophical Studies*, 182(7), 1975–2003. https://doi.org/10.1007/s11098-025-02301-3

Kingsbury Barry, A. (2026a). Why AI Governance Frameworks Cannot Deploy: A Structural Analysis of the Execution-First Problem and the Incentive Inversion That Prevents It from Being Solved. *Niti Logic Working Paper*. https://doi.org/10.5281/zenodo.19410642

Kingsbury Barry, A. (2026b). The Governance Window: Voluntary Adoption, Mandate Dynamics, and the Irreversibility of Architecture Choice. *Niti Logic Working Paper*. https://doi.org/10.5281/zenodo.19433302

Kingsbury Barry, A. (2026c). Representational Sequestration and Discursive Admissibility: A Structural Analysis of Non-Deployable AI Governance Frameworks in Professional Network Discourse. *Niti Logic Working Paper*. https://doi.org/10.5281/zenodo.19503567

Kingsbury Barry, A., and Montanez, F. (2026a). Adversarial Sufficiency: A Structural Analysis of Governance Failure and a New Standard for Assurance. *Niti Logic Working Paper*. https://doi.org/10.5281/zenodo.19583197

Kingsbury Barry, A., and Montanez, F. (2026b). Governance Without Consent: Why Compliant Systems Still Fail the Humans They Were Built to Serve. *Niti Logic Working Paper*. https://doi.org/10.5281/zenodo.19686648

Kulveit, J., Douglas, R., Ammann, N., Turan, D., Krueger, D., and Duvenaud, D. (2025). Gradual Disempowerment: Systemic Existential Risks from Incremental AI Development. arXiv:2501.16946 [cs.CY]. https://doi.org/10.48550/arXiv.2501.16946

Kwon, J., and Casper, S. (2026). Internal Deployment Gaps in AI Regulation. arXiv:2601.08005.

Mehrizi, M. H. R., et al. (2023). The Impact of AI Suggestions on Radiologists' Decisions: A Pilot Study of Explainability and Attitudinal Priming Interventions in Mammography Examination. *Scientific Reports*, 13, 9230. https://doi.org/10.1038/s41598-023-36435-3

Meyer, J. W., and Rowan, B. (1977). Institutionalized Organizations: Formal Structure as Myth and Ceremony. *American Journal of Sociology*, 83(2), 340–363.

Moon, A., and Boudreaux, B. (2026). *A Formal Model of How Artificial Intelligence Erodes Human Agency* (Report No. RR-A4817-1). RAND Corporation. https://doi.org/10.7249/RRA4817-1

Rosbach, E., et al. (2026). Stuck on Suggestions: Automation Bias, the Anchoring Effect, and the Factors That Shape Them in Computational Pathology. *Machine Learning for Biomedical Imaging*, 3, 126–147. https://doi.org/10.59275/j.melba.2026-87b1

Schembera, S., Haack, P., and Scherer, A. G. (2023). From Compliance to Progress: A Sensemaking Perspective on the Governance of Corruption. *Organization Science*, 34(3), 1184–1215. https://doi.org/10.1287/orsc.2022.1614

Simon, H. A. (1947). *Administrative Behavior: A Study of Decision-Making Processes in Administrative Organization*. Macmillan.

Wang, J., Huang, K., Klyman, K., and Bommasani, R. (2025). Do AI Companies Make Good on Voluntary Commitments to the White House? *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 8(3), 2625–2636. https://doi.org/10.1609/aies.v8i3.36743

© 2026 Thomas W. Gantz / Synthience Institute
License: CC-BY 4.0
synthience.org